

Sum-of-Squares and Non-Sum-of-Squares Regimes for the Toeplitz Böttcher–Wenzel Form

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Abstract

Toeplitz matrices are among the simplest structured models of translation invariance, arising naturally in convolution, stationary covariance models, and discretizations of translation-invariant operators. For a pair of real matrices, the Böttcher–Wenzel form (BW form) is a nonnegative quartic measuring the gap in the corresponding commutator inequality. László conjectured that, when both matrices are Toeplitz, this quartic is a sum of squares for every matrix order. We show that the conjecture is false. More precisely, we identify an explicit family of corner-supported Toeplitz pairs, retaining only the outermost diagonals on the two sides, for which the associated BW form is not a sum of squares at all sufficiently large orders. The proof reduces the complete family of Gram representations to a finite complex test: its Gram-invariant part develops a strictly negative limiting direction, while the remaining Gram-dependent contribution vanishes. We also give an explicit sufficient threshold for the matrix order. The negative result is complemented by two positive regimes. If either Toeplitz factor is symmetric or skew-symmetric, then the corresponding BW form is SoS for every order, with the other factor allowed to be an arbitrary real Toeplitz matrix. For unrestricted Toeplitz pairs, we further prove SoS representability for every $2 \leq N \leq 50$, with formal Lean 4 verification through order 20.

1 Introduction

Toeplitz matrices are among the simplest matrix models of translation invariance. Their entries are constant along each diagonal, so a matrix of order N is described by only $2N - 1$ parameters. A real Toeplitz matrix has constant entries along each diagonal. For $x = (x_{-(N-1)}, \dots, x_0, \dots, x_{N-1}) \in \mathbb{R}^{2N-1}$, write

$$X := (x_{i-j})_{i,j=1}^N = \begin{pmatrix} x_0 & x_{-1} & \cdots & x_{-(N-1)} \\ x_1 & x_0 & \ddots & \vdots \\ \vdots & \ddots & \ddots & x_{-1} \\ x_{N-1} & \cdots & x_1 & x_0 \end{pmatrix}. \quad (1)$$

They arise naturally in finite convolution, stationary covariance models, and the discretization of translation-invariant operators, random matrix theory, and structured optimization [5, 13, 2].

Let X and Y be Toeplitz matrices as in (1). It is well known that the Böttcher–Wenzel form (BW form) is a nonnegative homogeneous quartic polynomial with the $4N - 2$ real variables $(x_{-(N-1)}, \dots, x_{N-1}, y_{-(N-1)}, \dots, y_{N-1})$ [1, 9],

$$F_N(x, y) := 2\|X\|_F^2\|Y\|_F^2 - 2\langle X, Y \rangle_F^2 - \|[X, Y]\|_F^2 \quad (\text{BW form})$$

where $\langle X, Y \rangle_F = \text{tr}(X^\top Y)$, $\|X\|_F^2 = \langle X, X \rangle_F$ and $[X, Y] = XY - YX$ ¹.

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¹Indeed, for $X \neq 0$, apply the Böttcher–Wenzel inequality $\|[U, V]\|_F^2 \leq 2\|U\|_F^2\|V\|_F^2$ to $U = X$ and $V = Y - \langle X, Y \rangle_F X / \|X\|_F^2$; then $[U, V] = [X, Y]$ and $\|X\|_F^2\|V\|_F^2 = \|X\|_F^2\|Y\|_F^2 - \langle X, Y \rangle_F^2$, which gives $F_N(x, y) \geq 0$. The case $X = 0$ is immediate.

A more interesting question is whether this nonnegativity admits a sum-of-squares (SOS) representation $F_N(x, y) = \sum_{j=1}^r q_j(x, y)^2$, where r is finite and each q_j is a real homogeneous quadratic polynomial. Hilbert’s classical work established that nonnegative quartic forms need not be sums of squares in general [6, 3]. However, the homogeneous quartic in (BW form) is far from arbitrary: it has strong symmetries, arises from a matrix commutator, and is nonnegative at every order.

László conjectured in 2012 that such a sum-of-squares (SoS) representation exists for every $N \geq 2$ [9, Conjecture 15].

In 2012, László established SoS representability for $N \leq 7$ by expressing the form in wedge coordinates and adjusting its Gram matrix using Plücker identities [9, Section 5].

The conjecture was subsequently discussed by Lu and Wenzel in 2017 [10] and by Ge, Li, Tang and Zhou in their 2024 survey [4]; it is also listed as MI-15 in the OpenProblemsInNLA catalogue [11]. More recently, Holden supplied rational Gram certificates for $N = 8, \dots, 12$, extending the conjectured positive range to $N = 12$ [7]. Also, a related positive result holds for cyclic Hankel matrices, whose entries depend on $i + j$ modulo N : László proved that their Böttcher–Wenzel form is SoS at every order [9, Theorem 7]. For general Toeplitz pairs, however, László’s conjecture for the Toeplitz Böttcher–Wenzel form has remained open despite substantial positive evidence. The central structural question, in the spirit of Hilbert’s work, therefore remains:

Is Toeplitz structure sufficient to close the gap between nonnegativity and sums of squares for this family of quartic forms?

In this paper, we answer László’s conjecture in the negative: for every sufficiently large N , the Toeplitz Böttcher–Wenzel form F_N is not SoS. The main contribution of the paper is threefold.

The Toeplitz Böttcher–Wenzel form is not always a sum of squares. The proof isolates an explicit corner-supported subclass of Toeplitz pairs and reduces all positive semidefinite Gram representations to a common normal form. A finite complex test then separates a fixed Gram-invariant contribution from the remaining Gram-dependent part. The former has a strictly negative limiting direction, while the latter vanishes uniformly over all positive semidefinite Gram representations. This contradiction proves that the Toeplitz Böttcher–Wenzel form is not SoS for all sufficiently large orders. At the explicit sufficient threshold, the negative theorem is also formally verified in Lean 4; see Appendix F.3.

Symmetry subclasses remain SoS at every order. Although László’s conjecture is false in general, we establish a broad positive result valid for all N :

If either X or Y is symmetric or skew-symmetric, then the corresponding restriction of F_N is SoS at every order, with the other factor allowed to be any real Toeplitz matrix.

To the best of our knowledge, these explicit one-sided Toeplitz SoS representations have not previously appeared in the literature. These subclasses include several familiar matrix models. For instance, covariance matrices of real weakly stationary processes are symmetric Toeplitz [5], whereas odd convolution filters and centered first-difference stencils give skew-symmetric Toeplitz matrices. In the finite Fourier symbol $f_x(\theta) = \sum_{k=-(N-1)}^{N-1} x_k e^{ik\theta}$, the symmetric and skew-symmetric conditions correspond, respectively, to cosine terms and purely imaginary sine terms. Our result therefore provides explicit SoS representations for the Böttcher–Wenzel form on these subclasses at every order.

The Toeplitz Böttcher–Wenzel form remains SoS at small orders. We establish SoS representability for every $2 \leq N \leq 20$, with proofs checked in Lean 4, extending the previously certified range $2 \leq N \leq 12$. We further extend the positive range to $2 \leq N \leq 50$ through an exact chain of rational identities, in which each form is obtained from its predecessor by a positive scaling, a linear substitution, and the addition of an SoS polynomial.

The remainder of the paper is organized as follows. Section 2, Section 3, and Section 4 establish the three contributions described above, respectively. Section 5 concludes the paper and discusses the remaining open questions. The appendices collect the supporting technical material, including the complete wedge-Gram characterization, the exact negative witness, the explicit large-order bound, the detailed positive identities, and the verification procedures and code instructions for the finite-order results.

2 Failure of a sum-of-squares representation for Toeplitz BW Form

In this section, we prove the following theorem.

Theorem 2.1. *There exists an integer N_0 such that, for every integer $N \geq N_0$, the polynomial F_N in (BW form) is nonnegative but is not a finite sum of squares of real homogeneous quadratic polynomials. An explicit sufficient value of N_0 is given in Section D.*

This section is organized as follows. In Section 2.1, following [9, Section 2], we introduce the wedge coordinates and the associated wedge Gram representation. These coordinates will be used throughout the paper. In Section 2.2, we introduce a class of corner-supported Toeplitz matrices. The Böttcher–Wenzel forms generated by this class will provide the non-SoS examples at sufficiently large matrix size.

Our proof proceeds in three steps. We first define the corresponding corner-supported Böttcher–Wenzel form, denoted by $\mathcal{F}_m$, and construct a convenient Gram representation for it (Section 2.2.1). Since Gram representations are not unique, we then use the symmetries of $\mathcal{F}_m$ to reduce the full Gram family to a general normal form (Lemma 2.4). Finally, we construct a finite matrix test which, for sufficiently large m , rules out positive semidefiniteness for every possible Gram representation of $\mathcal{F}_m$ (Section 2.2.2). This proves that $\mathcal{F}_m$ is not SoS for sufficiently large m , and hence, that F_N is not SoS for all sufficiently large N .

2.1 Wedge-coordinate and wedge Gram representation

Following [9, Section 2], we use the wedge-coordinate representation of (BW form) throughout the paper. The BW form of Toeplitz matrices X, Y is a nonnegative homogeneous quartic polynomial in the $4N - 2$ real variables

$$(x_{-(N-1)}, \dots, x_{N-1}, y_{-(N-1)}, \dots, y_{N-1}) \in \mathbb{R}^{4N-2} \quad (2)$$

with $x, y \in \mathbb{R}^{2N-1}$. We define the wedge coordinates of (x, y) by

$$z_{ab} = x_a y_b - x_b y_a, \quad -(N-1) \leq a < b \leq N-1, \quad (3)$$

where x_a denotes the entry of x indexed by a and y_b denotes the entry of y indexed by b . Collect these coordinates, in

$$z := (z_{-(N-1), -(N-2)}, z_{-(N-1), -(N-3)}, \dots, z_{-(N-1), N-1}, z_{-(N-2), -(N-3)}, \dots, z_{N-2, N-1})^\top \in \mathbb{R}^{\binom{2N-1}{2}}. \quad (4)$$

The vector z is a quadratic lifting of the original variables. The wedge coordinates should be viewed as a structured quadratic lifting of the original Toeplitz parameters in (2) rather than as a linear change of variables. Each Toeplitz matrix is determined by $2N - 1$ real parameters, so the pair (X, Y) is described by $4N - 2$ real variables. The wedge vector $z = (z_{ab})_{a < b}$, $z_{ab} = x_a y_b - x_b y_a$, has $\binom{2N-1}{2}$ coordinates, each of which is quadratic in the original variables. Thus the map $(x, y) \mapsto z = x \wedge y$ lifts the quartic form $F_N(x, y)$ to a quadratic form in the wedge coordinates, $F_N(x, y) = z^\top Q z$. The wedge representation is substantially smaller than the full degree-two monomial representation in the $4N - 2$ original variables, whose dimension is $\binom{4N-2}{2}$. The wedge coordinates are not algebraically independent; their quadratic relations are precisely the Plücker relations described in Lemma 2.1. Thus the wedge coordinates form a structured subcollection of the usual degree-two monomial representation rather than an invertible change of coordinates.

We call the real symmetric matrix $Q \in \mathbb{S}^{\binom{2N-1}{2}}$ a wedge Gram matrix of F_N if

$$F_N(x, y) = z^\top Q z.$$

Such a Gram matrix is generally not unique; Lemma 2.1 characterizes this full Gram family. It is well known that a polynomial is SoS if and only if it admits a positive semidefinite Gram representation [12, 8]. For the Böttcher–Wenzel form, Lemma 2.1 also shows that every quadratic factor in an SoS representation lies in the wedge span. Consequently,

$$F_N(x, y) \text{ is SoS} \iff F_N(x, y) = z^\top Q z \text{ for some } Q \succeq 0.$$

Lemma 2.1 (Complete characterization of wedge Gram representations). *Let I be a finite ordered set. Let $F(x, y)$ be a real homogeneous quartic satisfying*

$$F(x, 0) = F(0, y) = F(x, x) = 0 \quad \text{for all } x, y \in \mathbb{R}^I. \quad (5)$$

If F is a sum of squares, every quadratic factor is a linear combination of the z_{ab} . Moreover, the real symmetric corrections $\mathcal{E}$ satisfying $z^\top \mathcal{E} z \equiv 0$ are spanned by the matrices P_{abcd} specified by²

$$z^\top P_{abcd} z = 2(z_{ab}z_{cd} - z_{ac}z_{bd} + z_{ad}z_{bc}), \quad a < b < c < d. \quad (6)$$

Consequently, if $F = z^\top G z$, then

$$F \text{ is SoS} \iff G + \mathcal{E} \succeq 0 \text{ for some } \mathcal{E} \in \text{span}\{P_{abcd}\}. \quad (7)$$

Proof. Write $F = \sum_j q_j^2$. Evaluating at $y = 0$ shows that every $q_j(x, 0)$ vanishes identically, and evaluating at $x = 0$ gives the analogous conclusion. Thus $q_j = x^\top A_j y$ for a real matrix A_j . The identity $q_j(x, x) = 0$ implies $A_j + A_j^\top = 0$, whence $q_j = \sum_{a < b} (A_j)_{ab} z_{ab}$. The coefficient vectors of the q_j give a positive semidefinite Gram matrix. To determine the relation space, group products of wedges by their multisets of labels. Monomials belonging to different multisets cannot cancel. For a multiset with a repeated label, there is at most one nonzero wedge product up to sign, and that product is not the zero polynomial. For four distinct labels, the three possible products are $z_{ab}z_{cd}$, $z_{ac}z_{bd}$ and $z_{ad}z_{bc}$. Any two are linearly independent, as is seen by comparing their coefficients, whereas their alternating combination in (6) vanishes. Their relation space is therefore exactly one-dimensional. This proves the asserted description of the correction space. Any representing Gram matrix differs from G by an element of this space. Conversely, a positive semidefinite representative factors as $R^\top R$, and the rows of Rz provide the required quadratic factors. This proves (7). $\square$

Note that Lemma 2.1 does not require Toeplitz structure. It applied to the Toeplitz Böttcher–Wenzel form since for the polynomial F_N , the three vanishing conditions (5) hold.

2.2 Toeplitz subclass that is non-SoS for large N

We begin by identifying a Toeplitz subclass on which the large-order failure of SoS representability will be detected. Fix a positive integer m and $N \geq 2m$. We call a Toeplitz matrix *corner-supported at depth m* if only the outermost m diagonals on each side are retained and all Toeplitz coefficients in the central band are set to zero. The following small example illustrates the geometry. Take $N = 6$ and $m = 2$, a corner-supported Toeplitz matrix with depth 2 has the form

$$X_{(N=6, m=2)} = \begin{pmatrix} 0 & 0 & 0 & 0 & x_{-4} & x_{-5} \\ 0 & 0 & 0 & 0 & 0 & x_{-4} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ x_4 & 0 & 0 & 0 & 0 & 0 \\ x_5 & x_4 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (8)$$

²Precisely, its entries at (ab, cd) , (ac, bd) , (ad, bc) are 1, -1 , 1, with the same entries at transposed positions and zero elsewhere.

Note that this example is only to illustrate the corner geometry. At this small depth $m = 2$, the corresponding BW form is still SoS; the failure of SoS representability occurs only for sufficiently large corner depth.

More generally, let X and Y be corner-supported Toeplitz matrices with at depth m , then we denote the indices of the retained entry (non-zero band) of Toeplitz labels as

$$\Lambda_{N,m} := \{\pm(N-1), \pm(N-2), \dots, \pm(N-m)\}$$

such that $x_a = y_a = 0$ for $a \notin \Lambda_{N,m}$. The (BW form) with respect to in corner-supported X, Y is the restriction to the the quartic

$$\mathcal{F}_m := F_N(x, y) \Big|_{x_a=y_a=0 \ (a \notin \Lambda_{N,m})}. \quad (9)$$

Lemma 2.2 (Properties of the corner-supported Toeplitz BW form). *Let X and Y be corner-supported Toeplitz matrices of size N and depth m , with $N \geq 2m$, and let $\mathcal{F}_m$ be the restricted Böttcher–Wenzel form defined in (9). Let $z = (z_{ab})_{a < b}$, $z_{ab} := x_a y_b - x_b y_a$, be the full wedge vector defined in (3), and let*

$$z_r := (z_{ab})_{a,b \in \Lambda_{N,m}, a < b} \in \mathbb{R}^{\binom{2m}{2}} \quad (10)$$

be its restriction to the retained labels. Then the following hold.

1. If $F_N = z^\top G z$ is any Gram representation in the full wedge coordinates, then

$$\mathcal{F}_m = z_r^\top G_r z_r,$$

where G_r is the principal submatrix of G indexed by the retained wedge coordinates.

2. After identifying the retained labels by their side and depth $p = 0, \dots, m-1$, the polynomial $\mathcal{F}_m$ depends only on m and is independent of N .

3. If $\mathcal{F}_m$ is not SoS, then F_N is not SoS for every $N \geq 2m$.

Proof. For the first statement, setting $x_a = y_a = 0$ for $a \notin \Lambda_{N,m}$ annihilates every wedge containing a non-retained label. Hence $z^\top G z$ restricts directly to $z_r^\top G_r z_r$.

For the second statement, let U_a be the shift matrix $(U_a)_{ij} = \mathbf{1}_{i-j=a}$, $0 \leq i, j < N$. Distinct shifts are Frobenius orthogonal and $\|U_a\|_F^2 = N - |a|$. The weighted Lagrange identity [9, Lemma 10] and bilinearity of the commutator give

$$F_N = 2 \sum_{a < b} (N - |a|)(N - |b|) z_{ab}^2 - \left\| \sum_{a < b} z_{ab} [U_a, U_b] \right\|_F^2. \quad (11)$$

For a retained label, $N - |\pm(N-1-a)| = a+1$. Shifts on the same side commute, while for $C_{ab} := [U_{-(N-1-a)}, U_{N-1-b}]$ one has, for $N \geq 2m$,

$$\langle C_{ab}, C_{cd} \rangle_F = 2 \mathbf{1}_{a-b=c-d} \min(a+1, b+1, c+1, d+1). \quad (12)$$

Thus every coefficient of the restricted quartic depends only on the depths a, b, c, d , and hence only on m .

Finally, linear restriction preserves sums of squares. Therefore an SoS representation of F_N would restrict to an SoS representation of $\mathcal{F}_m$, proving the last statement by contraposition. $\square$

2.2.1 Gram representation of the corner-supported Toeplitz BW form

In this subsection, we study the Gram representations of the corner-supported Toeplitz Böttcher–Wenzel form $\mathcal{B}$. We first construct one fixed Gram representation with a particularly simple block structure, and then derive a general normal form that captures all positive semidefinite Gram representations under the reduced restricted wedge coordinates.

To obtain the structured Gram representation, we first rearrange the restricted wedge vector as $z_r = (u^\top, v^\top, w^\top)^\top$, where $u, v \in \mathbb{R}^{\binom{m}{2}}$ and $w \in \mathbb{R}^{m^2}$. The entries are defined by

$$u_{ab} := x_a y_b - x_b y_a, \quad a < b < 0, \quad v_{ab} := x_a y_b - x_b y_a, \quad 0 < a < b, \quad w_{ab} := x_a y_b - x_b y_a, \quad a < 0 < b.$$

We refer to u and v as the two *pure wedge* families, since the two labels in each wedge lie on the same side of the main diagonal, and to w as the *mixed wedge* family, since its two labels lie on opposite sides. With this ordering, the restricted Gram representation takes a highly regular block form. The structure is already visible in the case $m = 2$.

Example 2.1 (Gram representation for a corner-supported Toeplitz BW form with $m = 2$). In (8), with the ordering described above, the corresponding restricted wedge vector and baseline Gram matrix are

$$z_r = \begin{pmatrix} u_{-5,-4} \\ v_{4,5} \\ w_{-5,5} \\ w_{-5,4} \\ w_{-4,5} \\ w_{-4,4} \end{pmatrix} \quad \left. \begin{array}{l} \left. \begin{array}{l} - \text{pure} \\ + \text{pure} \end{array} \right\} \\ \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} \text{mixed} \end{array} \right\} \quad G_2 = \begin{pmatrix} 4 & 2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & -2 & 0 \\ 0 & 0 & 0 & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 4 \end{pmatrix}.$$

Here the mixed coordinates are ordered by corner depth, $(w_{00}, w_{01}, w_{10}, w_{11}) = (w_{-5,5}, w_{-5,4}, w_{-4,5}, w_{-4,4})$. Thus,

$$\mathcal{B}_2 = z_r^\top G_2 z_r = 2(u_{-5,-4} + v_{4,5})^2 + 2u_{-5,-4}^2 + 2v_{4,5}^2 + 2(w_{-5,4} - w_{-4,5})^2 + 4w_{-4,4}^2.$$

Example 2.1 shows that this ordering arranges the restricted Gram matrix into a particularly simple block structure. The next lemma proves that the same structure holds for every corner depth m . For notational simplicity, in the rest of the paper, we index these three families $u_{ab} := u_{-(N-1-a), -(N-1-b)}$, $v_{ab} := v_{N-1-b, N-1-a}$, and $w_{ab} := w_{-(N-1-a), N-1-b}$. This is only a relabelling of the same restricted wedge coordinates.

Lemma 2.3 (Gram representation of the corner-supported Toeplitz BW form). Let $\mathcal{F}_m$ be a corner-supported Toeplitz Böttcher–Wenzel form generated by two Toeplitz matrices X, Y of size N and corner depth m , where $N \geq 2m$. Collect the coordinates into $z_r = (u^\top, v^\top, w^\top)^\top$, using lexicographic order within each depth-indexed group. Then

$$\mathcal{F}_m = z_r^\top G_m z_r, \quad G_m = \begin{pmatrix} D & K & 0 \\ K & D & 0 \\ 0 & 0 & B \end{pmatrix}. \quad (13)$$

Here $D, K \in \mathbb{S}^{\binom{m}{2}}$, $B \in \mathbb{S}^{m^2}$, and $G_m \in \mathbb{S}^{m(2m-1)}$ where $\mathbb{S}$ denotes the space of real symmetric matrices and the specific entries are

$$\begin{aligned} (D)_{ab,cd} &= 2(a+1)(b+1)\delta_{ac}\delta_{bd}, & (K)_{ab,cd} &= 2\mathbf{1}_{b-a=d-c} \min(a+1, c+1), \\ (B)_{ab,cd} &= 2(a+1)(b+1)\delta_{ac}\delta_{bd} - 2\mathbf{1}_{a+b=c+d} \min(a+1, b+1, c+1, d+1). \end{aligned} \quad (14)$$

For D, K , the index pairs satisfy $0 \leq a < b < m$ and $0 \leq c < d < m$; for B , all four indices range independently over $\{0, \dots, m-1\}$. In particular, D, K, B , and hence G_m , are fixed by (14) and are independent of N .

Proof. We rearrange the restricted expression (11) using the Plücker relations, as in the Gram construction of [9, Sections 2 and 5]. The raw Gram couples mixed wedges with equal differences $a - b = c - d$. For $a < c$ and $b < d$ satisfying this equality, the corresponding term is $-4 \min(a+1, b+1) w_{ab} w_{cd}$. With the orientation above, the Plücker identity becomes

$$u_{ac} v_{bd} + w_{ab} w_{cd} - w_{ad} w_{cb} = 0, \quad (15)$$

so that the mixed product is replaced by $-u_{ac} v_{bd} + w_{ad} w_{cb}$. Summing the resulting contributions gives (14). $\square$

The Gram representation in Lemma 2.3 is not unique. By Lemma 2.1, every other representation is obtained by adding Plücker corrections. The full Gram family is large, but the symmetries of $\mathcal{F}_m$ allow us to average any positive semidefinite representative into a simple normal form.

Lemma 2.4 (General Gram normal form for the corner-supported Toeplitz BW form). *Let $\mathcal{F}_m$ be a corner-supported Toeplitz Böttcher–Wenzel form generated by two Toeplitz matrices X, Y of size N and corner depth m , where $N \geq 2m$. Then the following statements hold.*

- The Gram representation of $\mathcal{F}_m$ is not unique. If $\mathcal{F}_m$ admits a positive semidefinite Gram representation, then it admits a symmetry-reduced representative of the form

$$Q_{\text{av}} = G_m + \begin{pmatrix} \mathcal{E}_0 & \mathcal{E}_1 & 0 \\ \mathcal{E}_1 & \mathcal{E}_0 & 0 \\ 0 & 0 & \mathcal{R}\mathcal{E}_1 \end{pmatrix} = \begin{pmatrix} D + \mathcal{E}_0 & K + \mathcal{E}_1 & 0 \\ K + \mathcal{E}_1 & D + \mathcal{E}_0 & 0 \\ 0 & 0 & B + \mathcal{R}\mathcal{E}_1 \end{pmatrix} \succeq 0, \quad (16)$$

where $\mathcal{E}_0, \mathcal{E}_1 \in \mathbb{S}^{\binom{m}{2}}$, $\mathcal{E}_0$ is a same-side Plücker correction acting within each pure wedge family (equivalently, a pure four-form), $\mathcal{E}_1$ is a real symmetric matrix coupling the two pure wedge families, and $\mathcal{R}$ is the realignment map induced by the corresponding index re-pairing.³

- Under the orthogonal change of coordinates $(u, v, w) \mapsto ((u+v)/\sqrt{2}, (u-v)/\sqrt{2}, w)$, the Gram representation in (16) becomes block diagonal:

$$\widehat{Q}_{\text{av}} := U^\top Q_{\text{av}} U = \begin{pmatrix} P_+ & 0 & 0 \\ 0 & P_- & 0 \\ 0 & 0 & M \end{pmatrix}, \quad (\text{Block-diag Gram})$$

where U is the orthogonal matrix representing the change of coordinates and

$$P_\pm = D + \mathcal{E}_0 \pm K \pm \mathcal{E}_1 \in \mathbb{S}^{\binom{m}{2}}, \quad M = B + \mathcal{R}\mathcal{E}_1 \in \mathbb{S}^{m^2}. \quad (17)$$

- The corner-supported form satisfies

$$\mathcal{F}_m \text{ is SoS} \iff \text{there exist } \mathcal{E}_0, \mathcal{E}_1 \text{ such that } Q_{\text{av}} \succeq 0 \iff P_+ \succeq 0, \quad P_- \succeq 0, \quad M \succeq 0.$$

Proof. The three statements follow from the complete Plücker description of the Gram freedom, the symmetries of $\mathcal{F}_m$, and the orthogonal decomposition of the two pure wedge families. A detailed proof is given in Section A. $\square$

2.2.2 A finite Gram test

We now reduce the search over the nonunique Gram family to a finite-dimensional matrix test; we denote the matrix as $\mathcal{G}_m$. The purpose of the construction is to separate the test matrix into a part depending only on the fixed polynomial data and a remaining part depending on the choice of Gram representation (i.e., $\mathcal{G}_m = \mathbf{L}_m - \mathbf{R}_m$). We will show that the first part has a negative limiting direction, whereas the second part is too small to compensate for it.

Lemma 2.5 (Finite Gram test). *Let $V \in \mathbb{C}^{m(2m-1) \times s}$ be an arbitrary complex matrix and let $\mathcal{G}_m$ be a complex Hermitian⁴ $s \times s$ matrix defined as*

$$\mathcal{G}_m := V^* \widehat{Q}_{\text{av}} V \in \mathbb{H}^s. \quad (18)$$

If $\mathcal{F}_m$ is SoS, then $\mathcal{G}_m \succeq 0$ for every V . In particular, $c^ \mathcal{G}_m c \geq 0$ for every $c \in \mathbb{C}^s$.*

³The full correction matrix is an element $\mathcal{E}$ of the space characterized in Lemma 2.1; $\mathcal{E}_0$ and $\mathcal{E}_1$ are the components remaining after the symmetry reduction. The realignment operator $\mathcal{R} : \mathbb{S}^{\binom{m}{2}} \rightarrow \mathbb{S}^{m^2}$ is a linear map that re-pairs the four indices of a pure-wedge Gram block into the corresponding mixed-wedge indices. In particular, it maps a correction on the pure wedge space to the induced correction on the mixed wedge space. The induced mixed correction $\mathcal{R}\mathcal{E}_1$ belongs to $\mathbb{S}^{m^2}$. See Section A for the detailed construction.

⁴ $\mathbb{H}^s := \{A \in \mathbb{C}^{s \times s} : A^* = A\}$ represents the space of complex Hermitian $s \times s$ matrices.

Proof. If $\mathcal{F}_m$ is SoS, Lemma 2.4 gives a normal-form Gram representation with $\hat{Q}_{\text{av}} \succeq 0$. Hence, for every $\hat{y} \in \mathbb{C}^s$,

$$\hat{y}^* \mathcal{G}_m \hat{y} = \hat{y}^* V^* \hat{Q}_{\text{av}} V \hat{y} = (V \hat{y})^* \hat{Q}_{\text{av}} (V \hat{y}) \geq 0.$$

Thus $\mathcal{G}_m \succeq 0$, and the last assertion follows by taking $\hat{y} = c$. $\square$

Remark 2.1 (Complex test points). *The original SoS problem and all Gram representations above remain over the real numbers. Complex numbers are introduced only in the test points used below, and do not change the underlying real SoS problem.*

Indeed, let Q be a real symmetric positive semidefinite matrix. For any complex vector $z = \hat{u} + i\hat{v}$, with $\hat{u}, \hat{v}$ real, we have

$$z^* Q z = (\hat{u}^\top - i\hat{v}^\top) Q (\hat{u} + i\hat{v}) = \hat{u}^\top Q \hat{u} + \hat{v}^\top Q \hat{v} \geq 0,$$

where the cross terms cancel because Q is symmetric. Thus a real positive semidefinite Gram matrix remains positive semidefinite when tested against complex vectors.

We may therefore choose several complex test points and arrange their corresponding feature vectors as the columns of a complex test matrix $\mathcal{V}$. The resulting matrix $\mathcal{V}^* Q \mathcal{V}$ must still be positive semidefinite whenever $Q \succeq 0$. Allowing complex test points gives us a larger and more flexible space of test directions, without enlarging the class of admissible Gram representations. In the construction in Section 2.2.3, we will choose complex points that approach $(i, -i)$ along prescribed directions as m increases. The associated test matrix develops a strictly negative limiting direction c . Since any real SoS Gram representation would remain positive semidefinite under the same complex test, this eventually yields the contradiction used to prove that the corner-supported form is not SoS.

Lemma 2.6 (Finite Gram decomposition). *Assume V contains a special block structure, namely*

$$V = \begin{pmatrix} V_1^*/\sqrt{2} & V_1^*/\sqrt{2} & V_3^* \end{pmatrix}^* \in \mathbb{C}^{m(2m-1) \times s}, \quad (19)$$

where $V_1 \in \mathbb{C}^{\binom{m}{2} \times s}$ and $V_3 \in \mathbb{C}^{m^2 \times s}$. Then, the finite Gram matrix in (18) admits the decomposition

$$\mathcal{G}_m = \mathbf{L}_m - \mathbf{R}_m \quad (20)$$

$$= \underbrace{V_1^* D V_1 + V_3^* (B + \mathcal{R}(D - K)) V_3}_{:= \mathbf{L}_m \text{ Gram-invariant}} - \underbrace{V_3^* (\mathcal{R} P_-) V_3}_{:= \mathbf{R}_m \text{ Gram-dependent}}. \quad (21)$$

Here, Gram-invariant means that $\mathbf{L}_m$ depends only on the fixed polynomial data D, K, B and is independent of the particular Gram representation, whereas Gram-dependent means that $\mathbf{R}_m$ depends on the chosen Gram representative through P_- .

Proof. From the block diagonal form of $\hat{Q}_{\text{av}}$,

$$\mathcal{G}_m = \begin{pmatrix} V_1^*/\sqrt{2} & V_1^*/\sqrt{2} & V_3^* \end{pmatrix} \begin{pmatrix} P_+ & 0 & 0 \\ 0 & P_- & 0 \\ 0 & 0 & M \end{pmatrix} \begin{pmatrix} V_1/\sqrt{2} \\ V_1/\sqrt{2} \\ V_3 \end{pmatrix} = V_1^* \frac{P_+ + P_-}{2} V_1 + V_3^* M V_3.$$

Using (17), $(P_+ + P_-)/2 = D + \mathcal{E}_0$ and $M = B + \mathcal{R}\mathcal{E}_1$. Moreover, $P_- = D - K + \mathcal{E}_0 - \mathcal{E}_1$ and $\mathcal{R}\mathcal{E}_0 = -\mathcal{E}_0$ ⁵. At the level of the corresponding finite evaluations, this gives $V_1^* \mathcal{E}_0 V_1 = -V_3^* (\mathcal{R}\mathcal{E}_0) V_3$. Hence

$$\begin{aligned} \mathcal{G}_m &= V_1^* D V_1 + V_3^* B V_3 - V_3^* \mathcal{R}(\mathcal{E}_0 - \mathcal{E}_1) V_3 \\ &= V_1^* D V_1 + V_3^* (B + \mathcal{R}(D - K)) V_3 - V_3^* (\mathcal{R} P_-) V_3, \end{aligned}$$

which proves (20). $\square$

The importance of Lemma 2.6 is that $\mathbf{L}_m$ depends only on the fixed matrices D, K, B , and hence is independent of the chosen Gram representation. The entire remaining dependence on the nonunique Gram representation is contained in $\mathbf{R}_m$. We will choose a finite family of points for which $\mathbf{L}_m$ develops a negative direction, while the contribution of $\mathbf{R}_m$ vanishes after the natural scaling.

⁵By definition, realignment exchanges the middle two indices, $(\mathcal{R}\mathcal{E}_0)_{ac,bd} = (\mathcal{E}_0)_{ab,cd} = \Omega_{abcd}$. Since Ω is totally antisymmetric, $\Omega_{acbd} = -\Omega_{abcd}$. Equivalently, $(\mathcal{R}\mathcal{E}_0)_{ac,bd} = -(\mathcal{E}_0)_{ac,bd}$. Hence $\mathcal{R}\mathcal{E}_0 = -\mathcal{E}_0$.

2.2.3 Choice of the finite test points

We now choose a specific family of test matrices V_m and a direction c for which the finite compression $\mathcal{G}_m = V_m^* \widehat{Q}_{\text{av}} V_m$ cannot remain positive semidefinite when m is sufficiently large.

Following the definition of V in (19). We next choose V so that the three diagonal blocks of $\widehat{Q}_{\text{av}}$ are tested on the following coordinates. For points $\mathbf{x}_\ell = (\xi_\ell, \eta_\ell) \in \mathbb{C}^2$, $1 \leq \ell \leq s$, define

$$V_1 := \left[(\xi_\ell^a \eta_\ell^b - \xi_\ell^b \eta_\ell^a)_{0 \leq a < b < m} \right]_{\ell=1}^s \in \mathbb{C}^{\binom{m}{2} \times s}, \quad V_3 := \left[(\xi_\ell^a \eta_\ell^b)_{0 \leq a, b < m} \right]_{\ell=1}^s \in \mathbb{C}^{m^2 \times s}, \quad (22)$$

For each m , define

$$\mathbf{x}_\ell^{(m)} = (\xi_\ell^{(m)}, \eta_\ell^{(m)}) := \left(i e^{-a_\ell / \sqrt{m}}, -i e^{-\bar{a}_\ell / \sqrt{m}} \right), \quad 1 \leq \ell \leq s. \quad (23)$$

where the parameters begin with

$$\begin{aligned} a_1 &= 1 + i, & a_2 &= 1 - i, & a_3 &= 2 + i, & a_4 &= 2 - i, & a_5 &= 1 + 2i, & a_6 &= 1 - 2i, \\ a_7 &= 2 + 2i, & a_8 &= 2 - 2i, & a_9 &= 1 + 3i, & a_{10} &= 1 - 3i, & a_{11} &= 2 + 3i, & a_{12} &= 2 - 3i, \dots \end{aligned}$$

and the pattern continues in conjugate pairs on the two vertical lines $\Re a = 1$ and $\Re a = 2$.

Note that $\mathbf{x}_\ell^{(m)} \rightarrow (i, -i)$ as $m \rightarrow \infty$. From now on $V_1, V_3, \mathbf{V}_m$ and $\mathbf{R}_m$ refer to the points in (23). The choice of $\Re a = 1, 2$ is not the only negative example. Additional families may also be included in the finite test (see Appendix C.2).

Proposition 2.1 (Negative limit of the Gram-invariant part). *Let $\mathbf{L}_m$ and $\mathbf{R}_m$ be defined as in Lemma 2.6. Fix the s test points in (23). Then the following statements hold.*

1. As $m \rightarrow \infty$, $\mathbf{x}_\ell^{(m)} \rightarrow (i, -i)$ and

$$\frac{1}{m^2} \mathbf{L}_m \longrightarrow \mathbf{L}_\star, \quad (24)$$

where $\mathbf{L}_\star \in \mathbb{S}^s$ is a fixed real symmetric $s \times s$ matrix, independent of m .

2. There exist a constant $\gamma > 0$ and an integer $m_0 > 0$ such that $\lambda_{\min}(\mathbf{L}_\star) < -\gamma < 0$. Let $c_\star \in \mathbb{R}^s$ be a corresponding real unit eigenvector. Since $\mathbf{L}_\star$ is independent of m , the same $c_\star$ is used for every $m \geq m_0$.
3. There exists an integer $m_1 \geq m_0$ such that, for every $m \geq m_1$, define $\tilde{c}_m := c_\star / m$, then

$$\tilde{c}_m^* \mathbf{L}_m \tilde{c}_m < -\frac{\gamma}{2} < 0. \quad (25)$$

4. If $\widehat{Q}_{\text{av}} \succeq 0$, there exists an integer $m_2 \geq m_0$ such that, for every $m \geq m_2$,

$$|\tilde{c}_m^* \mathbf{R}_m \tilde{c}_m| < \frac{\gamma}{4}, \quad (26)$$

uniformly over all positive semidefinite normal-form Gram representations.

Proof. We prove the four statements in order. For the first statement, the entries of $\mathbf{L}_m$ are explicit finite sums determined by D, K, B and the points $\mathbf{x}_\ell^{(m)}$. Since $\mathbf{x}_\ell^{(m)} \rightarrow (i, -i)$ at rate $m^{-1/2}$, the corresponding expressions have a leading term of order m^2 . For $1 \leq \ell, h \leq s$, define

$$A_{\ell h} := |a_\ell + \bar{a}_h|^2, \quad C_{\ell h} := 4 \Re(a_\ell) \Re(a_h).$$

Expanding the leading term gives

$$\frac{1}{m^2} (\mathbf{L}_m)_{\ell h} \longrightarrow 2 \left(\frac{2}{A_{\ell h}^2} + \frac{1}{C_{\ell h}^2} - \frac{1}{A_{\ell h} C_{\ell h}} \right) =: (\mathbf{L}_\star)_{\ell h}.$$

Thus $\mathbf{L}_\star := \left[2 \left(\frac{2}{A_{\ell h}^2} + \frac{1}{C_{\ell h}^2} - \frac{1}{A_{\ell h} C_{\ell h}} \right) \right]_{\ell, h=1}^s \in \mathbb{S}^s$ is a fixed real symmetric $s \times s$ matrix, independent of m . Since s is fixed, entrywise convergence implies convergence in any matrix norm, proving $\frac{1}{m^2} \mathbf{L}_m \longrightarrow \mathbf{L}_\star$.

For the second statement, the fixed two-family choice of test points determines $\mathbf{L}_\star$ completely. Appendix C gives an exact finite certificate showing that $\mathbf{L}_\star$ has a strictly negative eigenvalue. Hence there exists $\gamma > 0$ such that

$$\lambda_{\min}(\mathbf{L}_\star) < -\gamma < 0.$$

Let $c_\star \in \mathbb{R}^s$ be a corresponding real unit eigenvector: $\mathbf{L}_\star c_\star = \lambda_{\min}(\mathbf{L}_\star) c_\star$, $c_\star^\star c_\star = 1$. The matrix $\mathbf{L}_\star$, the constant γ , and the vector $c_\star$ are fixed and independent of m . Choose m_0 sufficiently large that the asymptotic estimates below are valid, and define $\tilde{c}_m := c_\star/m$ for every $m \geq m_0$.

For the third statement,

$$\tilde{c}_m^\star \mathbf{L}_m \tilde{c}_m = c_\star^\star \left(\frac{\mathbf{L}_m}{m^2} \right) c_\star \longrightarrow c_\star^\star \mathbf{L}_\star c_\star = \lambda_{\min}(\mathbf{L}_\star) < -\gamma.$$

Therefore there exists an integer $m_1 \geq m_0$ such that, for every $m \geq m_1$, $\tilde{c}_m^\star \mathbf{L}_m \tilde{c}_m < -\frac{\gamma}{2} < 0$.

For the fourth statement, suppose that $\hat{Q}_{\text{av}} \succeq 0$. Positivity of the mixed block first forces the low-order mixed directions into its kernel. The resulting improvement at the conjugate test points can then be continued to the realigned points appearing in $\mathbf{R}_m$. This yields

$$\sup_{\hat{Q}_{\text{av}} \succeq 0} \left| c_\star^\star \frac{\mathbf{R}_m}{m^2} c_\star \right| \longrightarrow 0.$$

The detailed argument is given in Appendix B. Hence there exists an integer $m_2 \geq m_0$ such that, for every $m \geq m_2$ and every positive semidefinite normal-form Gram representation, $|\tilde{c}_m^\star \mathbf{R}_m \tilde{c}_m| = |c_\star^\star \frac{\mathbf{R}_m}{m^2} c_\star| < \frac{\gamma}{4}$. This completes the proof. $\square$

Finally, we piece together the preceding lemmas to conclude the section and prove Theorem 2.1.

Proof of Theorem 2.1. Let $M_0 := \max\{m_1, m_2\}$ and suppose, for contradiction, that $\mathcal{F}_m$ is SoS for some $m \geq M_0$. By Lemma 2.5, the corresponding finite Gram test satisfies $\mathcal{G}_m \succeq 0$, and therefore $\tilde{c}_m^\star \mathcal{G}_m \tilde{c}_m \geq 0$. On the other hand, Lemma 2.6 and proposition 2.1 give

$$\tilde{c}_m^\star \mathcal{G}_m \tilde{c}_m \leq \tilde{c}_m^\star \mathbf{L}_m \tilde{c}_m + |\tilde{c}_m^\star \mathbf{R}_m \tilde{c}_m| < -\frac{\gamma}{2} + \frac{\gamma}{4} = -\frac{\gamma}{4} < 0,$$

contradicting $\mathcal{G}_m \succeq 0$. Therefore $\mathcal{F}_m$ is not SoS for every sufficiently large corner depth m . In particular, choose any integer $m_\star \geq M_0$. By Lemma 2.2, $\mathcal{F}_{m_\star}$ is a restriction of F_N for every $N \geq 2m_\star$. Hence F_N is not SoS for every $N \geq 2m_\star$. Taking $N_0 := 2m_\star$ proves the theorem. $\square$

3 Subclasses of Toeplitz BW Form that are SoS

The failure of SoS representability at large orders does not persist under every structural restriction. In this section, we show that the Toeplitz Böttcher–Wenzel form is SoS at every order whenever one factor is symmetric or skew-symmetric. The other factor remains an arbitrary real Toeplitz matrix.

The proof proceeds in two steps. First, Lemma 3.1 shows that the Toeplitz Böttcher–Wenzel form with one symmetric factor can be split into a term involving two symmetric Toeplitz matrices and a term involving a symmetric and a skew-symmetric matrix. Proposition 3.1 proves that the first term is SoS, while Lemma 3.2 proves the same for the second. The skew-symmetric case follows analogously, with the roles of the symmetric and skew-symmetric parts interchanged. After completing the proof, we discuss the exterior-product interpretation, and its relation to the two-matrix De Smet–Dillen–Verstraelen–Vrancken (DDVV) inequality. For notational simplicity, throughout this section we write $F(X, Y)$ for $F_N(x, y)$ in (BW form), where X and Y are the matrices of size N .

Splitting the factor Clearly, every real Toeplitz matrix X admits an orthogonal decomposition into a symmetric Toeplitz matrix and a skew-symmetric Toeplitz matrix:

$$X = S_x + K_x, \quad S_x = \frac{X + X^\top}{2}, \quad K_x = \frac{X - X^\top}{2}. \quad (27)$$

Here S_x is symmetric Toeplitz and K_x is skew-symmetric Toeplitz. Moreover, they are orthogonal with respect to the Frobenius inner product, $\langle S_x, K_x \rangle_F = 0$, and hence $\|X\|_F^2 = \|S_x\|_F^2 + \|K_x\|_F^2$. The following lemma shows that this decomposition also splits the Böttcher–Wenzel form.

Lemma 3.1 (The Splitting Lemma). *Let X be a real Toeplitz matrix, and let S and K be real symmetric and skew-symmetric matrices, respectively, of the same order. Then*

$$\begin{aligned} F(X, S) &= F(S, X) = F(S, S_x) + F(S, K_x), \\ F(X, K) &= F(K, X) = F(K, S_x) + F(K, K_x), \end{aligned} \quad (28)$$

where S_x and K_x are defined in (27).

Proof. Symmetric and skew-symmetric matrices are orthogonal with respect to the Frobenius inner product. In particular,

$$\langle S, K_x \rangle_F = 0, \quad \|S_x + K_x\|_F^2 = \|S_x\|_F^2 + \|K_x\|_F^2.$$

Moreover, $[S, S_x]$ is skew-symmetric, whereas $[S, K_x]$ is symmetric. Hence

$$\|[S, S_x + K_x]\|_F^2 = \|[S, S_x]\|_F^2 + \|[S, K_x]\|_F^2.$$

Substitution into the definition of F proves the first identity. For the second identity, use $\langle K, S_x \rangle_F = 0$ and the fact that $[K, S_x]$ is symmetric, while $[K, K_x]$ is skew-symmetric. The same argument then gives the second identity. $\square$

The two types of pairs The splitting lemma Lemma 3.1 reduces the theorem to two types of BW forms. Lemma 3.2 shows that the symmetric–skew-symmetric term in (28) is SoS. In fact, this result holds for any symmetric–skew-symmetric pair, even without the Toeplitz assumption. Proposition 3.1 shows that the same-symmetry term in (28) is also SoS.

Lemma 3.2 (Symmetric–skew-symmetric pair). *For every real symmetric matrix $S = (s_{ij})_{1 \leq i, j \leq N}$ and real skew-symmetric matrix $K = (k_{ij})_{1 \leq i, j \leq N}$ of order N ,*

$$F(S, K) = \|SK + KS\|_F^2 + 4 \sum_{\ell=1}^N \sum_{1 \leq i < j < k \leq N} (s_{i\ell}k_{jk} - s_{j\ell}k_{ik} + s_{k\ell}k_{ij})^2. \quad (29)$$

Proof. Since $\langle S, K \rangle_F = 0$, $F(S, K) = 2\|S\|_F^2\|K\|_F^2 - \|SK - KS\|_F^2$. For a vector $v \in \mathbb{R}^N$, the exterior-product Lagrange identity gives

$$\frac{1}{2}\|v\|^2\|K\|_F^2 - \|Kv\|^2 = \sum_{i < j < k} (v_i k_{jk} - v_j k_{ik} + v_k k_{ij})^2.$$

This identity follows directly by expansion. Apply it to each column of S and sum to obtain

$$2\|S\|_F^2\|K\|_F^2 = 4\|KS\|_F^2 + 4 \sum_{\ell=1}^N \sum_{i < j < k} (s_{i\ell}k_{jk} - s_{j\ell}k_{ik} + s_{k\ell}k_{ij})^2.$$

Finally, $(SK)^\top = -KS$, so $\|SK\|_F = \|KS\|_F$. The parallelogram identity therefore yields $4\|KS\|_F^2 - \|SK - KS\|_F^2 = \|SK + KS\|_F^2$. Combining the two identities proves (29). Every entry of $SK + KS$, and every expression inside the remaining squares, is a homogeneous quadratic polynomial in the independent entries of S and K . $\square$

Proposition 3.1 (Both symmetric or skew-symmetric pair). *For every $N \geq 2$, $F(X, Y)$ is SoS if X, Y are both symmetric Toeplitz or both skew-symmetric Toeplitz.*

Proof sketch. In either case, the matrices are determined by their nonnegative diagonal parameters. We use the wedge coordinates introduced in Section 2.1. Set $m = N - 1$ and reverse the nonzero diagonal labels by writing

$$\hat{z}_{ab} = z_{N-b, N-a}, \quad 1 \leq a < b \leq m.$$

This indexing replaces the diagonal-length weights $(N - a)(N - b)$ by ab . We rewrite the Toeplitz form as

$$F^{\text{sym}}(X, Y) = 4N \sum_{b=1}^{N-1} (N - b) z_{0b}^2 + 4\Psi_{N-1}(\hat{z}), \quad F^{\text{skew}}(X, Y) = 4\Psi_{N-1}(\hat{z}). \quad (30)$$

Note that the full calculation is given in Section E.1. Here $\Psi_m(\hat{z})$ is defined by

$$\Psi_m(\hat{z}) := 2 \sum_{1 \leq a < b \leq m} ab \hat{z}_{ab}^2 - \sum_{\delta=1}^{m-1} \sum_{\substack{k \geq 0 \\ 2k < m-\delta}} \left(\sum_{p=k+1}^{m-\delta-k} \hat{z}_{p, p+\delta} \right)^2. \quad (31)$$

Intuitively, the inner sum in (31) selects a window from the sequence $\hat{z}_{1, 1+\delta}, \dots, \hat{z}_{m-\delta, m}$, successively removing k terms from each end. All we need to prove next is that $\Psi_m(\hat{z}(x, y))$ is SoS. For example, when $m = 4$ and $\delta = 1$, the two windows give

$$(\hat{z}_{12} + \hat{z}_{23} + \hat{z}_{34})^2 \quad \text{and} \quad \hat{z}_{23}^2.$$

The essential observation is that the wedges satisfy the Plücker relations recalled in Section 2.1. These relations allow the cross terms in the negative window squares to be re-paired without changing the polynomial in x, y . For instance,

$$-2\hat{z}_{12}\hat{z}_{34} = -2\hat{z}_{13}\hat{z}_{24} + 2\hat{z}_{14}\hat{z}_{23}.$$

Applying the prescribed re-pairing rule gives three types of squares: window sums along a fixed index total, differences of wedges with the same gap, and individual wedges. Proposition E.1 gives the complete identity and its coefficients. The window and difference terms have nonnegative integer weights. After allocating the diagonal terms to these squares, the remaining coefficient of $\hat{z}_{p, p+\delta}^2$, denoted $s_\delta(p)$ in Section E.2, satisfies $s_\delta(p) \geq p^2 > 0$. This estimate ensures that completing the squares leaves no negative remainder.

Consequently, Ψ_m is a sum of squares of linear combinations of the wedges, after using their Plücker relations. Since each wedge is quadratic in x, y , this is an SoS representation in the original variables. Equation (30) proves both assertions. $\square$

Combining the splitting in Lemma 3.1 with the two results above gives the following all-order positive result.

Theorem 3.1 (One symmetric or skew-symmetric factor). *For every $N \geq 2$, the Toeplitz Böttcher–Wenzel form $F(X, Y)$ is SoS whenever either X or Y is symmetric or skew-symmetric Toeplitz. The other factor may be an arbitrary real Toeplitz matrix.*

Proof. Suppose first that X is symmetric Toeplitz. By Lemma 3.1, $F(X, Y) = F(X, S_y) + F(X, K_y)$, where the two terms are SoS by Proposition 3.1 and Lemma 3.2, respectively. The skew-symmetric case is identical with the two results interchanged. Finally, $F(X, Y) = F(Y, X)$ gives the same conclusion when the structural assumption is imposed on Y . $\square$

Remark 3.1. *The splitting above relies on one factor having a fixed symmetry. If both factors are decomposed into symmetric and skew-symmetric parts, additional cross terms appear and the four resulting BW forms no longer give the whole expression. Thus the argument proves the one-sided result above, but does not extend directly to unrestricted Toeplitz pairs. The full expansion is recorded in Appendix E.3.*

An exterior-product regularization viewpoint The positive part of the Böttcher–Wenzel form admits a useful regularization interpretation. By the weighted Lagrange identity,

$$\|X\|_F^2 \|Y\|_F^2 - \langle X, Y \rangle_F^2 = z^\top W_N z, \quad W_N \succ 0,$$

where $z = x \wedge y$ denotes the wedge coordinates introduced in Section 2.1. Since the commutator is linear in z , say $[X, Y] = \mathcal{C}_N z$, the generalized form

$$F_\tau(X, Y) := \tau(\|X\|_F^2 \|Y\|_F^2 - \langle X, Y \rangle_F^2) - \|[X, Y]\|_F^2$$

becomes

$$F_\tau(X, Y) = z^\top (\tau W_N - \mathcal{C}_N^* \mathcal{C}_N) z.$$

Thus, for each fixed N , the exterior-product term acts as a positive-definite regularizer in wedge space, and sufficiently large τ makes the lifted quadratic form positive semidefinite. Related interactions between regularization, global optimality, and SoS exactness for quartic polynomial models are studied in [16, 15, 14]. The Böttcher–Wenzel form corresponds to the fixed value $\tau = 2$. This differs from a radial regularization such as $\sigma\|(x, y)\|^4$: the exterior-product term vanishes whenever X and Y are linearly dependent, and is therefore degenerate in the original variables even though it is positive definite in wedge coordinates.

Relation to the two-matrix DDVV form A closely related commutator polynomial arises from the De Smet–Dillen–Verstraelen–Vrancken (DDVV) inequality for real symmetric matrices [10, 4]. For two such matrices, define

$$\mathcal{D}_N(X, Y) := (\|X\|_F^2 + \|Y\|_F^2)^2 - 2\|[X, Y]\|_F^2.$$

A direct expansion gives

$$\mathcal{D}_N(X, Y) = 2F(X, Y) + (\|X\|_F^2 - \|Y\|_F^2)^2 + 4\langle X, Y \rangle_F^2. \quad (32)$$

Hence an SoS representation of F immediately gives one for $\mathcal{D}_N$. The converse does not follow, since the last two terms in (32) are additional squares. The two-matrix DDVV form itself admits an SoS representation for arbitrary real symmetric matrices. Thus the DDVV result does not replace the Toeplitz argument in Proposition 3.1; rather, (32) shows precisely how the two quartic forms differ.

4 Exact certificates for the Toeplitz BW form at small orders

We have shown that, for sufficiently large N , the general Toeplitz Böttcher–Wenzel form is not SoS. In the previous section, we also showed that several structured subclasses remain SoS at every order.

This naturally raises the question of what happens for the small orders that arise in applications, when no additional symmetry structure is assumed. Is the unrestricted Toeplitz BW form still SoS? For small orders, the answer is positive. In this section, we show that the unrestricted Toeplitz BW form is SoS for every $2 \leq N \leq 50$. Moreover, for $2 \leq N \leq 20$, we provide a Lean-verified proof of the SoS property⁶.

Together with the large-order negative result, this leaves a substantial gap. The quantitative argument in Appendix B gives the explicit sufficient bound $N_0 = 2^{9961475}$ for non-SoS representability, whereas the exact positive construction below has currently been verified only through order 50. The latter stops because the positive increment used in the induction has only been constructed and verified through the step $n = 49$; this does not imply that F_{51} is not SoS. Thus the exact SoS status of the unrestricted Toeplitz BW form for $50 < N < N_0$ remains open. Determining the first non-SoS order, or substantially narrowing this gap, is a natural direction for future work.

Toeplitz BW SoS for $2 \leq N \leq 50$ We establish the result by an exact induction. Starting from F_2 , each step expresses the next Toeplitz BW form as a positive multiple of a linear substitution of the preceding form, plus an SoS increment.

Here N denotes the matrix order in the final result, while n is used only as the induction index. At the step from n to $n + 1$, let $z^{(n+1)}$ denote the wedge vector associated with the symbol variables of F_{n+1} .

⁶Note that the orders 21–50 are certified by exact arithmetic but are not included in the Lean verification.

Lemma 4.1 (Exact positive increments). *For each $n = 2, \dots, 49$, the supplied rational data specify a positive rational number α_n , a rational linear map $\psi_n : \mathbb{R}^{4n+2} \rightarrow \mathbb{R}^{4n-2}$, and a rational symmetric matrix $\mathbf{H}_n \succeq 0$ such that*

$$F_{n+1} = \alpha_n(F_n \circ \psi_n) + (z^{(n+1)})^\top \mathbf{H}_n z^{(n+1)}. \quad (33)$$

Proof. Each identity is verified by reconstructing the two quartics from their defining Toeplitz matrices and comparing all polynomial coefficients over $\mathbb{Q}$. Positive semidefiniteness of $\mathbf{H}_n$ is checked by exact rational matrix calculations after removing explicitly verified kernel directions. The data for all 48 steps and the verification procedure are given in Appendix F. $\square$

Theorem 4.1 (The finite positive range). *For every integer $2 \leq N \leq 50$, F_N is a sum of squares of real homogeneous quadratic polynomials. For $2 \leq N \leq 20$, the same statement is also formally verified in Lean 4.*

Proof. At order two, direct expansion gives $F_2 = 4z_{-1,0}^2 + 4z_{0,1}^2$. Suppose $F_n = \sum_j q_j^2$, where the q_j are real homogeneous quadratics. Since $\mathbf{H}_n \succeq 0$, it admits a real factorization $\mathbf{H}_n = V_n^\top V_n$. Equation (33) gives

$$F_{n+1} = \sum_j (\sqrt{\alpha_n} q_j \circ \psi_n)^2 + \sum_i (V_n z^{(n+1)})_i^2.$$

Each factor is again a homogeneous quadratic. Induction through $n = 49$ therefore proves the result for every $2 \leq N \leq 50$. $\square$

The substitutions in (33) allow the SoS form at one order to be reused at the next. For $2 \leq n \leq 16$, a simple rescaling of F_n is sufficient. For $17 \leq n \leq 49$, the substitution also damps the symbol at offset a by a rational factor $q_n^{|a|}$, with $0 < q_n \leq 1$, so that the remaining increment admits a positive semidefinite Gram representation. The precise constructions are given in Appendix F.

Lean verification through order 20. For $2 \leq N \leq 20$, the SoS assertion is also formally verified in Lean 4. In particular, the range $13 \leq N \leq 20$ extends the previously known positive range through order 12. The Lean formalization works with the original Toeplitz BW polynomial and proves directly that it is a finite sum of real homogeneous quadratic squares. The corresponding formal statement and source dependencies are recorded in Appendix F.4.

This verification is distinct from the exact rational induction in Section 4. The rational argument proves the SoS property for every $2 \leq N \leq 50$ by checking the identities (33) and the positive semidefiniteness of the associated rational matrices. The Lean proof, on the other hand, provides an independent formal verification for each order $2 \leq N \leq 20$ and need not use the same Gram representations. Thus,

orders	verification
$2 \leq N \leq 20$	exact rational verification and Lean 4
$21 \leq N \leq 50$	exact rational verification

Both results concern unrestricted real Toeplitz pairs. The status of the unrestricted form remains open for the intermediate range $50 < N < N_0$.

5 Conclusion

This paper shows that Toeplitz structure alone is not sufficient to guarantee a sum-of-squares representation of the Böttcher–Wenzel form. The main negative result is obtained by restricting the Toeplitz form to a corner-supported family whose structure is independent of the matrix order. After describing the complete family of Gram representations of this restricted form, we separate a fixed Gram-invariant contribution from the remaining Gram-dependent part. A finite complex test then detects a negative limiting direction in the former, while the latter vanishes uniformly over all positive semidefinite Gram representations. This gives the contradiction and proves that the unrestricted Toeplitz BW form is not SoS for every sufficiently large order.

At the same time, the failure is not universal across structured Toeplitz classes. We prove that if either factor is symmetric or skew-symmetric Toeplitz, then the BW form remains SoS at every order, with the other factor completely unrestricted within the real Toeplitz class. The proof follows from an orthogonal decomposition into symmetric and skew-symmetric parts, together with SoS identities for pairs of the same symmetry and for symmetric–skew-symmetric pairs. Thus the negative result is caused not by Toeplitz structure itself, but by the interaction between the two symmetry components when both factors contain them.

These two results leave several natural questions open. Most immediately, the smallest order at which the unrestricted Toeplitz BW form ceases to be SoS is still unknown. The present negative argument gives only a sufficient large-order threshold, while exact computations show that the form remains SoS throughout a substantial initial range. A second direction is to identify further structured subspaces, beyond the symmetric and skew-symmetric cases considered here, on which the interaction between the two symmetry components can still be controlled. More generally, it would be interesting to understand whether the corner and Gram-realignment mechanisms developed here can be used to locate sharper boundaries between nonnegativity and SoS in other structured polynomial matrix inequalities.

Declaration of generative AI and AI-assisted technologies During the preparation of this work the authors used large language model systems (Claude, Anthropic; Codex models, OpenAI) to generate candidate constructions and code, to run computations, to draft mathematical arguments, and to improve the readability of the text. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data availability The exact certificate data, verification scripts, and Lean 4 formalization are available at <https://github.com/erenup/toeplitz-bw-not-sos>. The archived release used for the results reported here is tag v1.0; the release documentation is recorded at commit 7126c08.

A Proof of the general Gram normal form

This appendix proves Lemma 2.4. We follow the three statements of the lemma in order.

Proof of Lemma 2.4. Put $r = \binom{m}{2}$.

1. Reduction of the Gram family. By Lemma 2.1, every Gram representation of $\mathcal{F}_m$ differs from the fixed representation

$$G_m = \begin{pmatrix} D & K & 0 \\ K & D & 0 \\ 0 & 0 & B \end{pmatrix}$$

by Plücker corrections. We classify these corrections according to how their four labels are distributed between the two sides of the Toeplitz corner.

Four labels on the same side produce corrections acting entirely within one pure wedge family. Such a correction has entries $(\mathcal{E}_0)_{ab,cd} = \Omega_{abcd}$, where Ω is totally antisymmetric in its four indices. These are the same-side Plücker corrections, or equivalently the pure four-forms.

Two labels on each side produce the Plücker relations that couple the two pure families with the mixed family. After the pure cross correction is collected into a symmetric matrix $\mathcal{E}_1 \in \mathbb{S}^r$, the corresponding change in the mixed block is determined by the same Plücker relations. We denote this induced matrix by $\mathcal{R}\mathcal{E}_1 \in \mathbb{S}^{m^2}$. Equivalently, after antisymmetrically extending the entries of $\mathcal{E}_1$ in each pure index pair, the realignment map is given by

$$(\mathcal{R}\mathcal{E}_1)_{ac,bd} = (\mathcal{E}_1)_{ab,cd}, \quad (\mathcal{R}\mathcal{E}_1)_{ad,bc} = -(\mathcal{E}_1)_{ab,cd}.$$

Thus $\mathcal{R}$ simply re-pairs the four indices from the two pure wedge pairs into two mixed wedge pairs.

Corrections involving three labels on one side and one label on the other produce pure–mixed blocks. To remove these terms, change the sign of all variables on the negative side. This transformation leaves the

pure coordinates u, v unchanged and changes the sign of every mixed coordinate w . The polynomial $\mathcal{F}_m$ is unchanged, so if $Q \succeq 0$ is a Gram representation, averaging Q with its image under this symmetry preserves both the represented polynomial and positive semidefiniteness, while eliminating all pure–mixed blocks.

Next interchange the two sides of the Toeplitz corner. Under this symmetry,

$$(u, v) \mapsto (-v, -u), \quad w_{ab} \mapsto -w_{ba}.$$

A second averaging therefore makes the two pure diagonal corrections equal and makes the pure cross correction symmetric. Consequently, whenever a positive semidefinite Gram representation exists, there is one of the form

$$Q_{\text{av}} = G_m + \begin{pmatrix} \mathcal{E}_0 & \mathcal{E}_1 & 0 \\ \mathcal{E}_1 & \mathcal{E}_0 & 0 \\ 0 & 0 & \mathcal{R}\mathcal{E}_1 \end{pmatrix},$$

which is exactly (16). Both averaging operations are convex averages of positive semidefinite matrices and therefore preserve positive semidefiniteness.

2. Block diagonalization. Write

$$A := D + \mathcal{E}_0, \quad C := K + \mathcal{E}_1.$$

Then the two pure blocks of Q_{av} have the form

$$\begin{pmatrix} A & C \\ C & A \end{pmatrix}.$$

Consider the orthogonal matrix

$$U = \begin{pmatrix} \frac{1}{\sqrt{2}}I_r & \frac{1}{\sqrt{2}}I_r & 0 \\ \frac{1}{\sqrt{2}}I_r & -\frac{1}{\sqrt{2}}I_r & 0 \\ 0 & 0 & I_{m^2} \end{pmatrix}.$$

It corresponds to the change of coordinates

$$(u, v, w) \mapsto \left(\frac{u+v}{\sqrt{2}}, \frac{u-v}{\sqrt{2}}, w \right).$$

Since $U^\top U = I$, direct multiplication gives

$$\begin{aligned} U^\top Q_{\text{av}} U &= \begin{pmatrix} A+C & 0 & 0 \\ 0 & A-C & 0 \\ 0 & 0 & B+\mathcal{R}\mathcal{E}_1 \end{pmatrix} \\ &= \begin{pmatrix} D+\mathcal{E}_0+K+\mathcal{E}_1 & 0 & 0 \\ 0 & D+\mathcal{E}_0-K-\mathcal{E}_1 & 0 \\ 0 & 0 & B+\mathcal{R}\mathcal{E}_1 \end{pmatrix}. \end{aligned}$$

Hence

$$P_+ = D + \mathcal{E}_0 + K + \mathcal{E}_1, \quad P_- = D + \mathcal{E}_0 - K - \mathcal{E}_1, \quad M = B + \mathcal{R}\mathcal{E}_1,$$

which proves (Block-diag Gram)–(17).

3. Equivalence with the SoS condition. Suppose first that $\mathcal{F}_m$ is SoS. By Lemma 2.1, it has a positive semidefinite Gram representation in the restricted wedge coordinates. Part 1 shows that this representation can be replaced, without losing positive semidefiniteness, by one of the form (16). Thus there exist $\mathcal{E}_0, \mathcal{E}_1$ for which $Q_{\text{av}} \succeq 0$.

Conversely, if such matrices $\mathcal{E}_0, \mathcal{E}_1$ exist and $Q_{\text{av}} \succeq 0$, then $Q_{\text{av}} = R^\top R$ for some real matrix R . Since Q_{av} is a Gram representation of $\mathcal{F}_m$,

$$\mathcal{F}_m = z_r^\top Q_{\text{av}} z_r = \|Rz_r\|^2,$$

which is a sum of squares of homogeneous quadratic polynomials. Therefore

$$\mathcal{F}_m \text{ is SoS} \iff \exists \mathcal{E}_0, \mathcal{E}_1 \text{ such that } Q_{\text{av}} \succeq 0.$$

Finally, U is orthogonal, so

$$Q_{\text{av}} \succeq 0 \iff U^\top Q_{\text{av}} U \succeq 0.$$

By the block diagonal form in Part 2, this is equivalent to

$$P_+ \succeq 0, \quad P_- \succeq 0, \quad M \succeq 0.$$

This proves the lemma. $\square$

B Qualitative estimates for the finite Gram test

This appendix supplies the qualitative estimates used in Proposition 2.1. The main theorem only requires their large- m limits. We also record, at the end of the appendix, the quantitative constants used to obtain the explicit sufficient order in Section D.

B.1 The scaled limit of the Gram-invariant part

Let

$$\delta_m := m^{-1/2}, \quad \mathbf{x}_\ell^{(m)} = (\xi_\ell^{(m)}, \eta_\ell^{(m)}) = (ie^{-\delta_m a_\ell}, -ie^{-\delta_m \bar{a}_\ell}), \quad 1 \leq \ell \leq s.$$

The speed parameters $a_1, \dots, a_s$ are fixed throughout, with $\Re a_\ell > 0$. Let $\mathbf{L}_{\infty, m}$ denote the matrix obtained from $\mathbf{L}_m$ by extending the depth indices in D, K, B from $\{0, \dots, m-1\}$ to all nonnegative integers.

For the pair (ℓ, h) , set

$$\begin{aligned} A_m^{\ell h} &:= (1 - \xi_\ell^{(m)} \overline{\xi_h^{(m)}})(1 - \eta_\ell^{(m)} \overline{\eta_h^{(m)}}), \\ C_m^{\ell h} &:= (1 - \xi_\ell^{(m)} \eta_\ell^{(m)})(1 - \overline{\xi_h^{(m)}} \overline{\eta_h^{(m)}}), \\ \Delta_m^{\ell h} &:= (1 - \xi_\ell^{(m)} \overline{\eta_h^{(m)}})(1 - \eta_\ell^{(m)} \overline{\xi_h^{(m)}}). \end{aligned}$$

Direct summation of the geometric series associated with D, K, B gives

$$(\mathbf{L}_{\infty, m})_{\ell h} = 2 \left(\frac{2}{(A_m^{\ell h})^2} + \frac{1}{(C_m^{\ell h})^2} - \frac{1}{A_m^{\ell h} C_m^{\ell h}} - \frac{2}{(\Delta_m^{\ell h})^2} \right). \quad (34)$$

Lemma B.1 (Finite-depth tail). *For the fixed finite set of test points, there exist constants $C, c > 0$, independent of m , such that*

$$\max_{1 \leq \ell, h \leq s} |(\mathbf{L}_m - \mathbf{L}_{\infty, m})_{\ell h}| \leq C m^4 e^{-c\sqrt{m}}.$$

Consequently,

$$\frac{1}{m^2} (\mathbf{L}_m - \mathbf{L}_{\infty, m}) \longrightarrow 0.$$

Proof. All coefficients in the finite sums defining D, K, B and their realignments grow at most polynomially, of degree four, in the depth indices. Since the set of speeds is fixed and $\rho_0 := \min_\ell \Re a_\ell > 0$, every test-point monomial of total depth q has modulus at most $e^{-\rho_0 q / \sqrt{m}}$. Every term discarded when the infinite sums are truncated at depth m contains at least one index of size at least m . Summing the resulting polynomially weighted geometric tails gives

$$C m^4 e^{-c\sqrt{m}}$$

for suitable constants $C, c > 0$. Division by m^2 proves the second assertion. $\square$

For the fixed speed parameters define

$$A_{\ell h} := |a_\ell + \bar{a}_h|^2, \quad C_{\ell h} := 4 \Re(a_\ell) \Re(a_h). \quad (35)$$

Since $1 - e^{-\delta z} = \delta z + O(\delta^2)$ as $\delta \rightarrow 0$, uniformly for the present finite set of z 's,

$$mA_m^{\ell h} \rightarrow A_{\ell h}, \quad mC_m^{\ell h} \rightarrow C_{\ell h}, \quad \Delta_m^{\ell h} \rightarrow 4.$$

Combining this with (34) and Lemma B.1 gives

$$\frac{1}{m^2}(\mathbf{L}_m)_{\ell h} \rightarrow 2 \left(\frac{2}{A_{\ell h}^2} + \frac{1}{C_{\ell h}^2} - \frac{1}{A_{\ell h}C_{\ell h}} \right).$$

Since s is fixed, the convergence is also convergence in every matrix norm. This is the limit matrix $\mathbf{L}_\star$ used in Proposition 2.1.

B.2 Qualitative control of the Gram-dependent part

We now prove the only uniform estimate needed in the main theorem: for the fixed finite test,

$$\sup_{\hat{Q}_{\text{av}} \succeq 0} \frac{1}{m^2} \max_{\ell, h} |(\mathbf{R}_m)_{\ell h}| \rightarrow 0.$$

The argument has two ingredients. Positivity first eliminates all low-total mixed modes. This gives an improved estimate on conjugate points. A qualitative maximum-principle argument then propagates the improvement to the finitely many realigned points. For $0 \leq g \leq 2m - 2$, let

$$s_g := \sum_{\substack{p+q=g \\ 0 \leq p, q < m}} \mathbf{e}_{pq}$$

in the mixed coordinates.

Lemma B.2 (Forced low mixed modes). *The baseline mixed block B is positive semidefinite, and every positive semidefinite normal-form Gram representation satisfies*

$$Ms_g = Bs_g = 0, \quad 0 \leq g < m. \quad (36)$$

Proof. The matrix B is block diagonal with respect to the total $g = p + q$. On a complete block, its off-diagonal entries are nonpositive and its row sums vanish; hence the block is a weighted graph Laplacian and is positive semidefinite. Incomplete blocks are principal submatrices, so $B \succeq 0$.

To obtain the second assertion, substitute the corner variables so that the pure wedges vanish and the mixed wedge w_{pq} equals t^{p+q} . If $Q \succeq 0$ is any feasible Gram matrix, the resulting polynomial is

$$\left\| \sum_g t^g Q^{1/2} s_g \right\|^2.$$

The same polynomial computed from the baseline mixed block has no term of degree below $2m$. If $g < m$ were the least index for which $Q^{1/2} s_g \neq 0$, the coefficient of t^{2g} would be strictly positive, a contradiction. Thus $Qs_g = 0$ for $g < m$. Restricting to the mixed block gives (36). $\square$

Write

$$\Psi_m := M - B.$$

By Lemma B.2, the repeated generating expression associated with Ψ_m contains only total degrees at least m in each pair of variables. At the conjugate points used below this high-order support is exponentially suppressed. Together with the explicit baseline block $D - K$, this yields the following qualitative estimate.

Lemma B.3 (Conjugate-point improvement). *Let α, β range over fixed compact real intervals with $\alpha > 0$, and set*

$$\xi_m = ie^{-(\alpha+i\beta)/\sqrt{m}}, \quad \eta_m = -ie^{-(\alpha-i\beta)/\sqrt{m}}.$$

Uniformly over all positive semidefinite normal-form Gram representations,

$$P_-[\xi_m, \eta_m] = O(m),$$

whereas on every fixed complex neighbourhood on which the two node moduli stay strictly inside the unit disk at distance comparable with $m^{-1/2}$,

$$P_-[\xi, \eta] = O(m^2).$$

Proof. The global estimate follows from positivity and the diagonal pure block:

$$0 \leq P_-[\xi, \eta] \leq \frac{8}{(1 - |\xi|^2)^2(1 - |\eta|^2)^2}.$$

At the scaled points above, the right-hand side is $O(m^2)$.

On the conjugate slice, the same-side four-form vanishes and the Plücker correction can be rewritten through Ψ_m . The baseline term $D - K$ is $O(m)$ there, while the high-total support of Ψ_m from Lemma B.2 contributes $o(m)$. Hence $P_-[\xi_m, \eta_m] = O(m)$ uniformly. A direct tail estimate proving the last $o(m)$ statement is obtained exactly as in Lemma B.1; the quantitative version is recorded below. $\square$

We use the following standard qualitative form of the two-constant principle.

Lemma B.4 (Qualitative continuation). *Let W_m be a holomorphic vector-valued function on a fixed rectangle $\Omega \subset \mathbb{C}$, possibly with a range dimension depending on m . Assume $\|W_m\| \leq C$ on Ω and $\|W_m\| \rightarrow 0$ uniformly on a nontrivial open segment I of one side of the rectangle. Then $\|W_m\| \rightarrow 0$ uniformly on every compact subset of Ω having positive harmonic measure with respect to I .*

Proof. For nonzero W_m , the function $\log \|W_m\|$ is subharmonic; zeros are handled by the usual value $-\infty$. The two-constant theorem bounds $\log \|W_m(z)\|$ by the boundary majorant weighted by the harmonic measure of I . On a fixed compact subset this harmonic measure is bounded below by a positive constant. Since the bound on I tends to zero while the global bound remains fixed, the resulting interior bound also tends to zero. The identically zero case is immediate. $\square$

Proposition B.1 (Uniform vanishing of the Gram-dependent term). *For the fixed test points of (23),*

$$\sup_{\hat{Q}_{\text{av}} \geq 0} \frac{1}{m^2} \max_{1 \leq \ell, h \leq s} |(\mathbf{R}_m)_{\ell h}| \rightarrow 0. \quad (37)$$

Consequently, for every fixed $c \in \mathbb{C}^s$,

$$\sup_{\hat{Q}_{\text{av}} \geq 0} \left| c^* \frac{\mathbf{R}_m}{m^2} c \right| \rightarrow 0.$$

Proof. Let $P_- = \Gamma^\top \Gamma$ and define

$$\mathbf{h}_m(\alpha, \beta) := \frac{1}{m} \Gamma f \left(i e^{-(\alpha + i\beta)/\sqrt{m}}, -i e^{-(\alpha - i\beta)/\sqrt{m}} \right).$$

By Lemma B.3, $\|\mathbf{h}_m\|$ is uniformly bounded on fixed complex rectangles, while it tends uniformly to zero on the real conjugate slice.

Apply Lemma B.4 first in the α variable and then in the β variable. No explicit harmonic-measure constant is needed because the test set is finite and fixed. For each pair (ℓ, h) , put

$$\alpha_{\ell h} := \frac{a_\ell + \overline{a_h}}{2}, \quad \beta_{\ell h} := \frac{a_\ell - \overline{a_h}}{2i}.$$

These finitely many points lie in a fixed compact set. Realignment gives

$$\frac{1}{m^2} (\mathbf{R}_m)_{\ell h} = \|\mathbf{h}_m(\alpha_{\ell h}, \beta_{\ell h})\|^2,$$

and therefore every entry tends to zero, uniformly over all feasible Gram representations. Since there are only finitely many entries, (37) follows. Finally,

$$\left| c^* \frac{\mathbf{R}_m}{m^2} c \right| \leq \|c\|_1^2 \frac{1}{m^2} \max_{\ell, h} |(\mathbf{R}_m)_{\ell h}|,$$

which proves the last assertion. $\square$

B.3 Explicit estimates for the sufficient threshold

For completeness, we now record the quantitative version of the preceding argument that is used in Section D. These constants are not needed for the qualitative proof in the main text.

For four arguments in the open unit disk define

$$\begin{aligned} A &= (1 - \xi\zeta)(1 - \eta\omega), & \Delta &= (1 - \xi\omega)(1 - \eta\zeta), \\ C &= (1 - \xi\eta)(1 - \zeta\omega), & \Pi &= \xi\eta\zeta\omega. \end{aligned} \quad (38)$$

At the level of scalar generating expressions, write $Q_{0,m} := D - K$ and let L_m denote the Gram-invariant kernel corresponding to $\mathbf{L}_m$. A subscript ∞ means that the depth indices are extended over all nonnegative integers. Direct summation of the geometric series gives

$$\begin{aligned} D_\infty &= 2(A^{-2} - \Delta^{-2}), \\ B_\infty &= 2(A^{-2} - (A\Delta)^{-1}), \\ Q_{0,\infty} &= D_\infty - \frac{2(\Delta - A)}{A\Delta C}, \\ L_\infty &= 2\left(\frac{2}{A^2} + \frac{1}{C^2} - \frac{1}{AC} - \frac{2}{\Delta^2}\right). \end{aligned} \quad (39)$$

Indeed, the first two formulas follow from $\sum_{p \geq 0} (p+1)t^p = (1-t)^{-2}$ and the common-shift identity for the minimum kernel. For the pure cross block, setting $q = p + h$, $s = r + h$, $h \geq 1$, gives

$$\sum_{p,r \geq 0} \min(p+1, r+1) (\xi\eta)^p (\zeta\omega)^r = \frac{1}{C(1-\Pi)},$$

and

$$\sum_{h \geq 1} (\eta^h - \xi^h)(\omega^h - \zeta^h) = \frac{(1-\Pi)(\Delta - A)}{A\Delta}.$$

Realignment interchanges A and C and leaves Δ unchanged, which yields the last line of (39).

Write $\mathcal{E}_{\text{mix}} := M - B$. By Lemma B.2, the repeated generating expression $\mathcal{E}_{\text{mix}}(\xi, \xi, \zeta, \zeta)$ contains only total degrees at least m in each pair of variables. Moreover, positivity and (17) imply the global bound

$$0 \leq P_\pm[\xi, \eta] \leq \frac{8}{(1 - |\xi|^2)^2(1 - |\eta|^2)^2}, \quad |\xi|, |\eta| < 1. \quad (40)$$

On conjugate nodes, the same-side four-form vanishes and

$$P_-[\xi, \bar{\xi}] = Q_{0,m}[\xi, \bar{\xi}] + \mathcal{E}_{\text{mix}}(\xi, \xi, \bar{\xi}, \bar{\xi}). \quad (41)$$

If $\xi = \sqrt{\rho}e^{i\vartheta}$, the corresponding infinite baseline is

$$Q_{0,\infty}[\xi, \bar{\xi}] = \frac{8\rho \sin^2 \vartheta}{(1 - \rho)^2((1 - \rho)^2 + 4\rho \sin^2 \vartheta)^2}. \quad (42)$$

Lemma B.5 (Uniform finite tails). *For $0 < \epsilon \leq 1$, set*

$$\mathcal{T}(m, \epsilon) = 2^{32} \epsilon^{-8} e^{-\epsilon m/4}. \quad (43)$$

If all four argument moduli are at most $e^{-\epsilon/2}$, then $|L_m - L_\infty| \leq \mathcal{T}(m, \epsilon)$. For every positive semidefinite normal-form Gram representation,

$$|P_-[\xi, \bar{\xi}] - Q_{0,\infty}[\xi, \bar{\xi}]| \leq \mathcal{T}(m, \epsilon) \quad (|\xi| \leq e^{-\epsilon/2}). \quad (44)$$

Proof. The diagonal of the realigned correction vanishes, so the positive semidefinite matrices M and B have the same diagonal. For $W = (p+1)(q+1)(r+1)(s+1)$, Cauchy–Schwarz gives

$$|(\mathcal{E}_{\text{mix}})_{pq,rs}| \leq 4\sqrt{W} \leq 4W.$$

The coefficients of the baseline kernels are bounded by a constant multiple of W . Every discarded finite-depth term contains at least one index of size at least m . Splitting the radial decay in half and using $1 - e^{-\epsilon/4} \geq \epsilon/8$ therefore gives

$$|L_m - L_\infty|, |Q_{0,m} - Q_{0,\infty}| \leq 2^{30} \epsilon^{-8} e^{-\epsilon m/4}.$$

For the correction in (41), Lemma B.2 leaves only total degrees at least m in both variable pairs; the same summation gives

$$|\mathcal{E}_{\text{mix}}(\xi, \xi, \bar{\xi}, \bar{\xi})| \leq 2^{26} \epsilon^{-8} e^{-\epsilon m/2}.$$

Combining the two estimates proves the lemma. $\square$

For the explicit threshold set

$$k \geq 16, \quad \epsilon = 2^{-k}, \quad m = \epsilon^{-2} = 2^{2k}. \quad (45)$$

Then

$$\mathcal{T} \leq 2^{32+8k-2^{k-2}}, \quad \mathcal{T} \leq \epsilon^{-2}, \quad \epsilon^4 \mathcal{T} \leq \epsilon. \quad (46)$$

Lemma B.6 (Quantitative continuation). *Let $P_- = \Gamma^\top \Gamma$ be a real Gram factorization, define $\mathbf{h}(\xi, \eta) = \Gamma f(\xi, \eta)$, and set*

$$\mathbf{h}_\epsilon(\alpha, \beta) = \mathbf{h}(ie^{-\epsilon(\alpha+i\beta)}, -ie^{-\epsilon(\alpha-i\beta)}). \quad (47)$$

For the parameters in (45),

$$\epsilon^4 \|\mathbf{h}_\epsilon(\alpha, \beta)\|^2 \leq 2^{20} \epsilon^{2^{-17}} \quad (48)$$

throughout

$$1 \leq \Re \alpha \leq 2, \quad |\Im \alpha| \leq 256, \quad |\Re \beta| \leq 256, \quad |\Im \beta| \leq \frac{1}{2}. \quad (49)$$

The bound is uniform over the dimension of the Gram factorization.

Proof. From (40) and $1 - e^{-\epsilon/2} \geq \epsilon/4$,

$$\|\mathbf{h}_\epsilon(\alpha, \beta)\|^2 \leq 2^{20} \epsilon^{-4}$$

whenever $\Re \alpha - |\Im \beta| \geq 1/4$. On the real conjugate slice, (42), Lemma B.5, and (46) improve this to

$$\|\mathbf{h}_\epsilon(\alpha, \beta)\|^2 \leq 2^{20} \epsilon^{-2}$$

for $\alpha \in [1/2, 1024 + 1/2]$ and $\beta \in [-512, 512]$.

We apply the two-constant principle twice. On the upper α -rectangle $1/2 \leq \Re \alpha \leq 1024 + 1/2$, $0 \leq \Im \alpha \leq 512$, use

$$h_1(\alpha) = \sin \frac{\pi(\Re \alpha - 1/2)}{1024} \frac{\sinh(\pi(512 - \Im \alpha)/1024)}{\sinh(\pi/2)}.$$

On $1 \leq \Re \alpha \leq 2$ and $0 \leq \Im \alpha \leq 256$ one has $h_1 > 2^{-12} > 2^{-13}$. Repeating in the lower rectangle yields

$$\|\mathbf{h}_\epsilon(\alpha, \beta)\|^2 \leq 2^{20} \epsilon^{-4+2 \cdot 2^{-13}}$$

for $1 \leq \Re \alpha \leq 2$, $|\Im \alpha| \leq 256$, and real $\beta \in [-512, 512]$.

For fixed such α , apply the same argument in the upper and lower β -rectangles with

$$h_2(\beta) = \cos \frac{\pi \Re \beta}{1024} \frac{\sinh(\pi(3/4 - \Im \beta)/1024)}{\sinh(3\pi/4096)}.$$

On $|\Re \beta| \leq 256$ and $|\Im \beta| \leq 1/2$, its relevant harmonic measure is larger than 2^{-5} . Since $2 \cdot 2^{-13} \cdot 2^{-5} = 2^{-17}$, the second comparison gives (48). $\square$

C An exact negative certificate for $\mathbf{L}_\star$

This appendix certifies the only finite computation used in Proposition 2.1: the fixed matrix $\mathbf{L}_\star$ has a strictly negative eigenvalue. The construction uses exactly the two vertical families of test speeds listed in the main text.

For the certificate only, it is convenient to parameterize the same finite set by

$$I = \{(r, j, \sigma) : r \in \{1, 2\}, 1 \leq j \leq 256, \sigma \in \{-1, 1\}\},$$

with

$$a_{r,j,\sigma} = r + i\sigma j.$$

For $\ell = (r, j, \sigma)$ and $h = (r', j', \sigma')$, the entries of $\mathbf{L}_\star$ are

$$(\mathbf{L}_\star)_{\ell h} = 2 \left(\frac{2}{A_{\ell h}^2} + \frac{1}{C_{\ell h}^2} - \frac{1}{A_{\ell h} C_{\ell h}} \right), \quad (50)$$

where

$$A_{\ell h} = (r + r')^2 + (\sigma j - \sigma' j')^2, \quad C_{\ell h} = 4rr'.$$

Thus every entry is rational.

Define the integer vector $d \in \mathbb{Z}^{1024}$ by

$$d_{r,j,\sigma} := \sigma \widehat{d}_j, \quad \widehat{d}_j := \left\lfloor \frac{2^{43} j}{(128^2 + j^2)^2} \right\rfloor. \quad (51)$$

Lemma C.1 (Exact negative direction). *The vector in (51) satisfies*

$$\|d\|_1 = 859652168 < 2^{30}, \quad d^\top \mathbf{L}_\star d < -2^{42}. \quad (52)$$

Consequently,

$$\lambda_{\min}(\mathbf{L}_\star) < -2^{-18}. \quad (53)$$

In particular, the constant γ in Proposition 2.1 may be chosen as $\gamma = 2^{-18}$.

Proof. Since the entries in (50) are rational, the quadratic form can be checked using integer arithmetic only. Define

$$\mathcal{J} := \sum_{\ell, h \in I} \left\lfloor \frac{2^{64} d_\ell d_h (4C_{\ell h}^2 + 2A_{\ell h}^2 - 2A_{\ell h} C_{\ell h})}{A_{\ell h}^2 C_{\ell h}^2} \right\rfloor.$$

Each summand is rounded toward $-\infty$. Since $|I|^2 = 2^{20}$,

$$\frac{\mathcal{J}}{2^{64}} \leq d^\top \mathbf{L}_\star d \leq \frac{\mathcal{J} + 2^{20}}{2^{64}}.$$

The exact integer evaluation is

$$\begin{aligned} \mathcal{J} &= -111807741093765987869670624504700, \\ \mathcal{J} + 2^{20} &= -111807741093765987869670623456124, \end{aligned}$$

whose upper numerator is smaller than -2^{106} . Hence $d^\top \mathbf{L}_\star d < -2^{42}$. The stated ℓ_1 norm follows directly from (51).

Now

$$\|d\|_2^2 \leq \|d\|_1^2 < 2^{60}.$$

Therefore the Rayleigh quotient satisfies

$$\frac{d^\top \mathbf{L}_\star d}{\|d\|_2^2} < -\frac{2^{42}}{2^{60}} = -2^{-18}.$$

By the variational characterization of the smallest eigenvalue,

$$\lambda_{\min}(\mathbf{L}_\star) \leq \frac{d^\top \mathbf{L}_\star d}{\|d\|_2^2} < -2^{-18},$$

which proves (53). □

Remark C.1. A direct floating-point diagonalization gives $\lambda_{\min}(\mathbf{L}_\star) \approx -1.18428 \times 10^{-2}$, which is substantially more negative than the conservative exact margin in (53). The numerical value is not used in the proof.

Remark C.2 (Alternative families of test points). The two-family choice $\Re a = 1, 2$ is not unique. Preliminary numerical experiments with three vertical families

$$\Re a \in \{1, 2, 3\}, \quad \Im a = \pm 1, \dots, \pm J,$$

show a negative eigenvalue already near $J = 44$, corresponding to $s = 6J = 264$ test points. We retain the two-family construction because it is the family for which the exact certificate above is supplied.

D An explicit finite threshold

The main proof needs only the qualitative limits in Proposition B.1. Here we retain the explicit constants from the quantitative argument and obtain a sufficient finite order.

Theorem D.1 (Uniform finite-scale non-SoS test). Let k, ϵ, m satisfy (45). If $\mathcal{F}_m$ is SoS, then every positive semidefinite normal-form Gram representation (16) gives a nonnegative scalar Φ_m satisfying

$$0 \leq \Phi_m < -2^{42} + 2^{90}m^{-1/2} + 2^{80}m^{-1/262144}. \quad (54)$$

All constants are independent of the Gram corrections and of the rank of the representation.

Proof. Use the fixed data of (51), and put $\mathbf{x}_\ell = (\xi_\ell, \eta_\ell) = (ie^{-\epsilon a_\ell}, -ie^{-\epsilon \bar{a}_\ell})$. Then $\eta_\ell = \bar{\xi}_\ell$. Define

$$\mathcal{G}_m = \left[\left(M + \frac{P_+ + P_-}{2} \right) (\bar{\mathbf{x}}_\ell, \mathbf{x}_h) \right]_{\ell, h \in I}, \quad \Phi_m = \epsilon^4 d^* \mathcal{G}_m d. \quad (55)$$

Each summand is an evaluated positive semidefinite kernel on the same list of points. Taking the direct sum of the pure and mixed feature spaces proves $\mathcal{G}_m \succeq 0$, hence $\Phi_m \geq 0$.

Let $\mathbf{L}_m$ and $\mathbf{R}_m$ be the evaluation matrices of L_m and $\mathcal{R}P_-$ at these points. Realignment gives

$$(\mathcal{R}P_-)(\bar{\mathbf{x}}_\ell, \mathbf{x}_h) = P_- (\bar{\xi}_\ell, \xi_h, \xi_\ell, \bar{\xi}_h) = P_- [\bar{\xi}_\ell, \xi_h] = P_- [\xi_\ell, \bar{\xi}_h]. \quad (56)$$

The last equality uses real symmetry of P_- . For $\ell = (r, j, \sigma)$ and $h = (r', j', \sigma')$, take

$$\alpha = \frac{r + r'}{2} + i \frac{\sigma j - \sigma' j'}{2}, \quad \beta = \frac{\sigma j + \sigma' j'}{2} - i \frac{r - r'}{2}.$$

Then $\alpha + i\beta = a_\ell$ and $\alpha - i\beta = \bar{a}_h$. These parameters satisfy (49), so Lemma B.6 applies to every entry of (56). Consequently,

$$|\epsilon^4 d^* \mathbf{R}_m d| \leq \|d\|_1^2 2^{20} \epsilon^{2^{-17}} < 2^{80} \epsilon^{2^{-17}}. \quad (57)$$

It remains to compare the finite baseline with $\mathbf{L}_\star$. At the test points, each factor of A and C has the form $1 - e^{-\epsilon \chi}$, where $\Re \chi \geq 2$ and $|\chi| < 2^{10}$. The integral identity

$$\frac{1 - e^{-\epsilon \chi}}{\epsilon \chi} = \int_0^1 e^{-\epsilon \chi u} du$$

and $|e^{-\epsilon \chi u} - 1| \leq \epsilon |\chi| u$ give a relative error at most $2^{10} \epsilon$ for each factor. Thus $A/\epsilon^2 = A_0(1 + \delta_A)$ and $C/\epsilon^2 = C_0(1 + \delta_C)$, where $|\delta_A|, |\delta_C| \leq 2^{12} \epsilon \leq 1/16$.

For $|\delta| \leq 1/4$ we use $|(1 + \delta)^{-1} - 1| \leq 2|\delta|$ and $|(1 + \delta)^{-2} - 1| \leq 6|\delta|$. At our points $A_0, C_0 \geq 4$. Each factor of Δ is $1 + e^{-\epsilon \chi}$, and $|\Im(\epsilon \chi)| \leq 512\epsilon < 1$, so the exponential has positive real part and $|\Delta| \geq 1$. Writing $\delta = \max(|\delta_A|, |\delta_C|)$, the product of the two inverse factors has relative error at most 5δ . The three reciprocal terms in (39) therefore contribute at most $(3/2 + 3/4 + 5/8)\delta < 3\delta \leq 3 \cdot 2^{12} \epsilon$; the remaining Δ term contributes at most $4\epsilon^4$. In particular,

$$|\epsilon^4 L_\infty(\bar{\mathbf{x}}_\ell, \mathbf{x}_h) - \mathbf{L}_\star(\ell, h)| \leq 2^{20} \epsilon.$$

All four node moduli are at most $e^{-\epsilon}$, so Lemma B.5 and (46) add at most ϵ to this scaled error. We obtain

$$|\epsilon^4 L_m(\bar{\mathbf{x}}_\ell, \mathbf{x}_h) - \mathbf{L}_\star(\ell, h)| \leq 2^{30}\epsilon. \quad (58)$$

Pairing (20) with d , and using (52), (57) and (58), gives

$$\begin{aligned} 0 &\leq \Phi_m = \epsilon^4 d^* \mathbf{L}_m d - \epsilon^4 d^* \mathbf{R}_m d \\ &< -2^{42} + 2^{90}\epsilon + 2^{80}\epsilon^{2^{-17}} \\ &= -2^{42} + 2^{90}m^{-1/2} + 2^{80}m^{-1/262144}. \end{aligned}$$

Here both entrywise errors are multiplied by $\|d\|_1^2 < 2^{60}$, and $\epsilon = m^{-1/2}$. This proves (54). $\square$

Proposition D.1 (Explicit sufficient order). *Let*

$$k = 38 \cdot 2^{17} + 1 = 4980737, \quad m = 2^{2k} = 2^{9961474}.$$

Then $\mathcal{F}_m$ is not SoS. Consequently, F_N is not SoS for every

$$N \geq 2m = 2^{9961475}.$$

Proof. Convexity of $x \mapsto 2^{-x}$ gives $2^{-x} \leq 1 - x/2$ for $0 \leq x \leq 1$. The upper bound in (54) is therefore at most

$$-2^{42} + 2^{90-k} + 2^{42}2^{-2^{-17}} \leq -2^{24} + 2^{90-k} < 0.$$

This contradicts $\Phi_m \geq 0$, so $\mathcal{F}_m$ is not SoS. By Lemma 2.2, it is a linear restriction of F_N for every $N \geq 2m = 2^{9961475}$, and any SoS representation of such an F_N would restrict to one of $\mathcal{F}_m$. $\square$

The size of this threshold comes from the slow term $2^{80}m^{-1/262144}$ in (54). The witness itself is fixed; the large order is needed only to make the uniform continuation error smaller than its negative margin. This bound is sufficient and is not intended to identify the first non-SoS order.

E Explicit certificates for the positive subclasses

This appendix supplies the reduction and the coefficient formulas used in Proposition 3.1. We first compute the commutator, then rearrange its window terms by Plücker identities. All index sets are finite; a sum over an empty set is zero.

E.1 Reduction for pairs with the same symmetry

For symmetric Toeplitz matrices write $X_{ij} = x_{|i-j|}$ and $Y_{ij} = y_{|i-j|}$. For skew-symmetric Toeplitz matrices write $X_{ij} = \text{sgn}(i-j)x_{|i-j|}$ and similarly for Y , with $x_0 = y_0 = 0$. In both cases $z_{ab} = x_a y_b - x_b y_a$. In the symmetric case the diagonal multiplicities are $w_0 = N$ and $w_a = 2(N-a)$ for $a > 0$. The weighted Lagrange identity [9, Lemma 10] gives

$$2\|X\|_F^2\|Y\|_F^2 - 2\langle X, Y \rangle_F^2 = 4N \sum_{b=1}^{N-1} (N-b)z_{0b}^2 + 8 \sum_{1 \leq a < b < N} (N-a)(N-b)z_{ab}^2.$$

The same formula holds in the skew-symmetric case with its first sum equal to zero.

Use matrix indices $0, \dots, N-1$, set $m = N-1$, and fix $\delta = j-i > 0$. For a symmetric pair, splitting the summation index in $[X, Y]_{ij}$ at i and j gives

$$[X, Y]_{i, i+\delta} = \sum_{t=0}^i z_{t, t+\delta} - \sum_{t=0}^{m-\delta-i} z_{t, t+\delta}.$$

The terms with an index strictly between i and j cancel in pairs. For a skew-symmetric pair the displayed expression changes sign. Both commutators are skew-symmetric. Each nonempty window occurs twice above the diagonal, so

$$\| [X, Y] \|_F^2 = 4 \sum_{\delta=1}^{m-1} \sum_{\substack{k \geq 0 \\ 2k < m-\delta}} \left(\sum_{t=k+1}^{m-\delta-k} z_{t,t+\delta} \right)^2.$$

Reflection $t = N - \delta - p$ turns each window into the corresponding sum of $\hat{z}_{p,p+\delta} = z_{N-p-\delta, N-p}$. Combining these identities proves (30).

E.2 The window certificate

For $1 \leq \delta < m$, put $h_\delta = m - \delta$ and define

$$\kappa_\delta(p, q) = \min(p, q, h_\delta + 1 - p, h_\delta + 1 - q), \quad 1 \leq p, q \leq h_\delta.$$

This counts the nested intervals $[k+1, h_\delta - k]$ containing both p and q . Consequently,

$$\sum_{\substack{k \geq 0 \\ 2k < h_\delta}} \left(\sum_{p=k+1}^{h_\delta-k} v_p \right)^2 = \sum_{p,q=1}^{h_\delta} \kappa_\delta(p, q) v_p v_q. \quad (59)$$

Thus the uncorrected Gram of Ψ_m has diagonal $2p(p + \delta) - \kappa_\delta(p, p)$ and entries $-\kappa_\delta(p, q)$ between distinct wedges of gap δ .

For $3 \leq \sigma \leq 2m - 1$, set

$$\begin{aligned} I_\sigma &= \{a \in \mathbb{Z} : \max(1, \sigma - m) \leq a < \sigma/2\}, \\ c_\sigma &= \max(0, \sigma - m - 1), \quad L_\sigma = \min(\sigma, m + 1), \\ \mathcal{W}_k^\sigma &= \{a \in I_\sigma : c_\sigma + 1 + k \leq a \leq L_\sigma - 1 - k\}, \\ t_\sigma(a, b) &= \min(a - c_\sigma, b - c_\sigma, L_\sigma - a, L_\sigma - b). \end{aligned}$$

For $1 \leq p < q \leq h_\delta$ with $e = q - p \leq \delta$, define

$$\omega_\delta(p, q) = \begin{cases} \kappa_\delta(p, q) + \kappa_e(p, p + \delta), & e < \delta, \\ \kappa_\delta(p, q), & e = \delta. \end{cases}$$

The remaining diagonal coefficient is

$$s_\delta(p) = 2p(p + \delta) - \kappa_\delta(p, p) - t_{2p+\delta}(p, p) - \sum_{\substack{1 \leq q \leq h_\delta \\ 0 < |q-p| \leq \delta}} \omega_\delta(\min(p, q), \max(p, q)). \quad (60)$$

Proposition E.1 (An explicit window decomposition). *For every $m \geq 1$, the reflected wedges in (31) satisfy*

$$\Psi_m(\hat{z}) = \sum_{\sigma=3}^{2m-1} \sum_{k \geq 0} \left(\sum_{a \in \mathcal{W}_k^\sigma} \hat{z}_{a, \sigma-a} \right)^2 + \sum_{\delta=1}^{m-1} \sum_{\substack{1 \leq p < q \leq h_\delta \\ q-p \leq \delta}} \omega_\delta(p, q) (\hat{z}_{p,p+\delta} - \hat{z}_{q,q+\delta})^2 + \sum_{\delta=1}^{m-1} \sum_{p=1}^{h_\delta} s_\delta(p) \hat{z}_{p,p+\delta}^2. \quad (61)$$

All nonempty edge weights are positive integers, and $s_\delta(p) \geq p^2$. In particular, the right-hand side is SoS in the original symbol variables.

Proof. For $p \geq 1$, $1 \leq \alpha < \beta$ and $p + \alpha + \beta \leq m$, the Plücker identity is

$$R_{p,\alpha,\beta} := \hat{z}_{p,p+\alpha} \hat{z}_{p+\beta,p+\alpha+\beta} - \hat{z}_{p,p+\beta} \hat{z}_{p+\alpha,p+\alpha+\beta} + \hat{z}_{p,p+\alpha+\beta} \hat{z}_{p+\alpha,p+\beta} = 0.$$

Add $2\kappa_\alpha(p, p + \beta)R_{p,\alpha,\beta}$ to Ψ_m for every such triple. These additions preserve the polynomial. Their effect on the Gram entries is as follows. For two wedges of the same gap δ at positions $p < q$, write $e = q - p$. If $e > \delta$,

the correction cancels the entry $-\kappa_\delta(p, q)$. If $e < \delta$, it changes that entry to $-\kappa_\delta(p, q) - \kappa_e(p, p + \delta) = -\omega_\delta(p, q)$. If $e = \delta$, no four distinct indices are present and the entry remains $-\omega_\delta(p, q)$.

The third product in $R_{p,\alpha,\beta}$ couples wedges on the same antidiagonal. For $a < b$ in I_σ , its coefficient in the Gram is

$$\kappa_{b-a}(a, \sigma - b) = \min(a, \sigma - b, m + 1 - b, m + 1 + a - \sigma) = t_\sigma(a, b).$$

Counting the sets $\mathcal{W}_k^\sigma$ containing both a and b expresses these entries by the first sum in (61). The second sum gives the negative same-gap entries, and (60) makes the diagonals agree. All remaining entries are zero. This proves the identity.

It remains to check the signs. Each tent value in its range is a positive integer, so $\omega_\delta(p, q) \geq 1$. The bounds $\kappa_\delta(p, p) \leq p$ and $t_{2p+\delta}(p, p) \leq p$ are immediate. For a right neighbour at distance $e < \delta$ the edge weight is at most $2p$, and at distance $e = \delta$ it is at most p . For a left neighbour these bounds are $2(p - e)$ and $p - \delta$, respectively. Including missing boundary neighbours only increases the upper bound, whence

$$\sum_q \omega_\delta(\min(p, q), \max(p, q)) \leq \begin{cases} (2\delta - 1)p + p(p - 1), & p \leq \delta, \\ (4\delta - 2)p - \delta^2, & p > \delta. \end{cases}$$

Inserting these estimates into (60) gives

$$s_\delta(p) \geq \begin{cases} p^2, & p \leq \delta, \\ p^2 + (p - \delta)^2, & p > \delta. \end{cases}$$

The sums in the certificate are finite because the windows are eventually empty. Every wedge is quadratic in the original variables, so the displayed identity is a finite sum of homogeneous quadratic squares. Integer weights may equivalently be realized by repeating the corresponding squares. $\square$

E.3 The coupling when both factors have both parts

Let $X = S_1 + K_1$ and $Y = S_2 + K_2$, with S_1, S_2 symmetric and K_1, K_2 skew-symmetric. Expanding the two Frobenius-orthogonal parts of the commutator gives

$$F(X, Y) = F(S_1, S_2) + F(K_1, K_2) + F(S_1, K_2) + F(K_1, S_2) + \mathcal{C},$$

where

$$\mathcal{C} = -4\langle S_1, S_2 \rangle_F \langle K_1, K_2 \rangle_F - 2\langle [S_1, S_2], [K_1, K_2] \rangle_F - 2\langle [S_1, K_2], [K_1, S_2] \rangle_F.$$

For Toeplitz matrices the four displayed F terms have the certificates of Section 3. The term $\mathcal{C}$ vanishes if one factor has only one symmetry part, but is present in general. No sign or SoS assertion for $\mathcal{C}$ is needed in the one-sided proof.

F Data and formal verification

This appendix specifies the 48 exact positive-increment steps used in Lemma 4.1 and the exact tests that establish their validity. The accompanying files `verification/data/step_02.json.gz` through `verification/data/step_49.json.gz` contain rational coefficients and their indices. The manifest lists the steps in order and gives hashes of both compressed and uncompressed data.

F.1 The two types of increment

For $2 \leq n \leq 16$, the certificate represents

$$R_n = (n - 1)F_{n+1} - (n + 1)F_n \tag{62}$$

as an SoS. In (33) this gives $\alpha_n = (n+1)/(n-1)$, the coordinate projection for ψ_n , and the Gram of R_n divided by $n-1$ for $\mathbf{H}_n$. For $17 \leq n \leq 49$, it represents

$$I_n = F_{n+1} - F_n \circ \psi_n, \quad \psi_n(x, y) = ((q_n^{|a|} x_a)_{|a| < n}, (q_n^{|a|} y_a)_{|a| < n}), \quad (63)$$

where $q_n = 1 - \theta_n/n$ is rational and $0 < q_n \leq 1$. The rational number θ_n is stored in the corresponding data file. Here $\alpha_n = 1$. In particular, $q_{49} = 97/98$ at the final step.

The stored Gram blocks use noncentral wedges, with both labels nonzero. Central wedges are added as separate squares. For each $1 \leq a \leq n$, the coefficient of each of $z_{-a,0}^2$ and $z_{0,a}^2$ is

$$\begin{cases} 2(n+1)(a-1), & \text{in } R_n, \\ 2((n+1)(n+1-a) - q_n^{2a}n(n-a)), & \text{in } I_n. \end{cases}$$

These coefficients are nonnegative. Adjoining them produces a Gram on all wedges, as required in (33).

F.2 Exact polynomial and positivity checks

For each step, the checker expands F_n and F_{n+1} directly from the Toeplitz matrix definition. It also expands the proposed Gram expression and verifies equality of every monomial coefficient. The Plücker corrections are separately checked to represent zero. All coefficients in these comparisons are integers or rationals.

The full noncentral Gram has verified kernel vectors

$$h_g = \sum_{\substack{-n \leq a < 0 < b \leq n \\ b-a=g}} e_{ab}, \quad n+1 \leq g \leq 2n,$$

where e_{ab} is the coordinate vector for z_{ab} . Their supports are disjoint. For each h_g , delete the first coordinate in its support, in lexicographic order. The remaining coordinate vectors together with the h_g form a basis. Once the exact relations $\mathbf{H}_n h_g = 0$ have been checked, this change of basis gives a congruence

$$\mathbf{H}_n \sim \begin{pmatrix} \hat{H}_n & 0 \\ 0 & 0 \end{pmatrix},$$

where $\hat{H}_n$ is the retained principal submatrix. Consequently $\mathbf{H}_n \succeq 0$ follows from $\hat{H}_n \succeq 0$. No theorem about the kernel of all possible Grams is needed for this check of the supplied matrices.

Each connected block of $\hat{H}_n$ is checked independently. One available test is Sylvester's criterion using exact fraction-free elimination. The default checker uses an integer residual certificate. Write a rational block as A/d , with A integral and $d > 0$. For an integer matrix L and integer $s \geq 0$, it computes

$$E = 2^{2s}A - dLL^\top, \quad \delta = \min_i \left(E_{ii} - \sum_{j \neq i} |E_{ij}| \right). \quad (64)$$

If $\delta > 0$, symmetry and diagonal dominance give $E \succeq \delta I$, hence

$$\frac{A}{d} = 2^{-2s}LL^\top + \frac{E}{d2^{2s}} \succeq \frac{\delta}{d2^{2s}}I \succ 0.$$

The proposed L may be obtained numerically, but every equality and inequality in this acceptance test uses exact integers. The rational factorization coefficients of an SoS itself are not required: a positive semidefinite rational Gram admits the real factorization used in Section 4.

Run `python3 verification/positive.py` after installing the dependencies in `verification/requirements.txt`. The checker validates the complete chain, including the base case, coefficient identities, central squares, kernel relations, and positivity. It also rejects deliberately corrupted identities and Gram corrections. The expected final line is

PASS: F_N is SOS for 2 <= N <= 50; 48 exact steps; negative controls rejected

The compressed coefficients are included with the source archive; the mathematical acceptance conditions are (62)–(64).

F.3 Lean verification of the large-order negative result

The public repository also contains a Lean 4 formalization of the large-order negative theorem for the original Toeplitz BW polynomial. In particular, the declaration `ToeplitzSOS.Negative.sharpNegativeResolution` proves that the original real Toeplitz BW form is not SoS for every $N \geq 2^{9961475}$, where the SoS predicate allows an arbitrary finite number of homogeneous quadratic squares with arbitrary real coefficients. The source is `lean/ToeplitzSOS/Negative/Resolution.lean`. The verified formalization is frozen at repository tag `v1.0`; reproduction commands and the exact verified source tree are recorded in `VERIFICATION.md`.

F.4 The positive Lean statement

In the supplied formalization, `toeplitzBW n` is the polynomial defined by the original real Toeplitz matrices, and `IsSumSqHomQuad` asserts the existence of a finite sum of real homogeneous quadratic squares. In the namespace `ToeplitzSOS`, the range theorem is

```
theorem toeplitzBW_isSumSq_of_le_20
  (n : Nat) (h2 : 2 <= n) (h20 : n <= 20) :
  IsSumSqHomQuad (toeplitzBW n)
```

Its proof combines the explicit order-two identity with the fixed-order theorems for $n = 3, \dots, 20$. The source file in the public repository is `lean/ToeplitzSOS/Certificates/Range.lean`. The Lean and Mathlib versions are pinned by the repository release. From the `lean/` directory, the positive certificates can be checked with `./verify --certificates`. The rational chain at orders 21–50 is not part of this theorem.

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